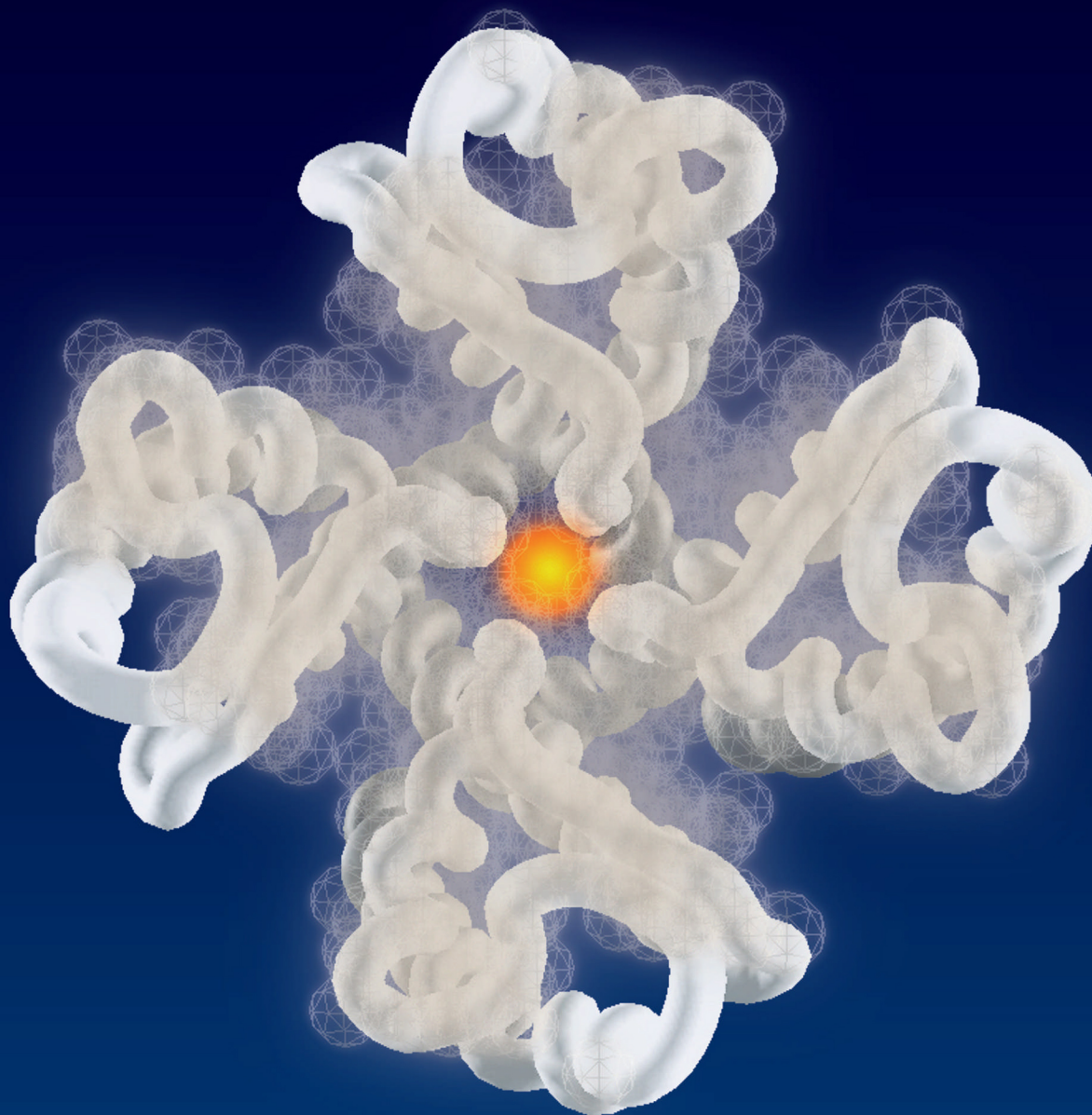


Introduction to Protein Structure

Second Edition



Carl Branden & John Tooze

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An aerial view of the European Synchrotron Radiation Facility at Grenoble, France, an advanced source of synchrotron x-ray radiation for use in the study of protein structure, as well as for use in the physical and material sciences. The synchrotron radiation is produced in the circular building in the lower left of the photograph. (Courtesy of ESRF.)

Introduction to Protein Structure

Second Edition

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THE COVER

Front: The structure of the potassium channel from *Streptomyces lividans*, determined by Roderick MacKinnon at the Rockefeller University, New York. As discussed in Chapter 12, this structure—the first of such an ion channel—shows how the channel allows the passage of potassium ions through cell membranes with high efficiency and selectivity. The view is looking down the protein as it sits in the cell membrane, as seen from outside the cell, with a potassium ion shown in gold. This image was produced using the GRASP program (A. Nicholls and B. Honig, Columbia University) from atomic coordinates kindly provided by Rodney MacKinnon.

Back: A hand-drawn image of the potassium channel, in the same view as on the front cover, with each subunit of the tetrameric protein shown in a different color.

Cover design by Christopher Thorpe and Nigel Orme.

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Preface

The determination of the atomic structures of proteins has seen an enormous increase in impetus since the first edition of this book was published in 1991. The number of new structures reported is close to doubling each year. Technical advances—for example the increased availability of synchrotron x-ray beams and methods for freezing crystals so as to reduce radiation damage to them, the development of multidimensional NMR and NMR machines with ever more powerful magnetic fields, and the exploitation of gene cloning, sequencing and expression systems have all contributed to the growth of protein structure determination. On the one hand, it is becoming increasingly easy to obtain relatively large amounts of naturally rare proteins, on the other hand the crystallographers can work with ever smaller crystals.

The fundamental tenet of molecular biology, namely that one cannot really understand biological reactions without understanding the structure of the participating molecules, is at last being vindicated. As the database of known protein structures rapidly expands, so does the range of biological pathways about which we can ask meaningful questions at close to atomic levels of resolution. An understanding of the principles of protein structure is becoming of ever widening significance to molecular biology.

The pharmaceutical industry has over the past decade become a major user of the protein structure databases, and a major contributor of newly determined structures. Knowledge of an enzyme's or a receptor's atomic structure is invaluable in the search for specific and strongly binding inhibitors. For example the quest for effective inhibitors of HIV protease, to be used in combination therapy for AIDS, led many pharmaceutical companies to determine the structure of that protease with bound inhibitors. Over 120 of these structure determinations have been done so far and at least two inhibitors of HIV protease are now being regularly used to treat AIDS. It seems certain that the determination of the atomic structure of target molecules will play an increasingly important role in drug design.

The commercial exploitation of our increased understanding of protein structure will not, of course, be restricted to the pharmaceutical industry. The industrial use of enzymes in the chemical industry, the development of new and more specific pesticides and herbicides, the modification of enzymes in order to change the composition of plant oils and plant carbohydrates are all examples of other commercial developments that depend, in part, on understanding the structure of particular proteins at high resolution.

As the complete genomes of more and more species are sequenced, the determination of the function of previously unidentified open reading frames is becoming an increasing and challenging problem. The possibility of

setting up centers for automated high through-put structure determinations is being seriously discussed. In the absence of any recognizable sequence homology to proteins of known function, this approach, surprising though it may seem, could become an effective way of determining function via structural homology.

The growth in the interest in high-resolution protein structure over the past decade and the reception of the first edition have encouraged us to prepare a new edition of this book. Universities are devoting more time to courses specifically on protein structure, or increasing the amount of time given to protein structure in more generally based biology and biochemistry courses. We hope that this new edition of *Introduction to Protein Structure* will prove useful both to teachers and students.

In 1988 when we began writing the first edition, about 250 protein structures had been determined to medium to high resolution and in those days a professional protein crystallographer was familiar with most of them. We were not therefore faced with a severe problem of what to leave out as we wrote. Today, the coordinates of over 6500 proteins have been deposited in the Protein Data Bank at Brookhaven, New York. Both the number of structures and the variety of biological systems to which they relate are so high that the field of protein structure is becoming more fragmented and specialized. It is becoming increasingly difficult to keep sight of the wood amongst so many trees. The question of what to include and what to omit is, for today's authors, crucially important. We have tried to resist the temptation to describe more and more proteins, adding detail but not increasing understanding of the basic concepts. This edition is inescapably a little larger than its predecessor, but to contain the increase in size we have deleted two chapters while adding four. We run the risk of disappointing not a few structural biologists whose favourite proteins are not mentioned. To them we apologize and ask for their understanding.

Acknowledgements

In preparing the second edition of this book we have again relied heavily on and benefited greatly from the advice and constructive criticism of numerous colleagues. We are particularly grateful to Ken Holmes (Max-Planck Institute, Heidelberg), Lawrence Stern (MIT), Michelle Arkin (Sunesis Pharmaceuticals), and Watson Fuller (Keele University, UK) for their contributions to, respectively, Chapters 14 and 18, Chapter 15, Chapter 17 and Chapter 18. Stephen Harrison (Harvard University) and Paul Sigler (Yale University) provided extensive help and advice on Chapters 8–10, 13 and 16, and Chapters 8–10 and 13, respectively, for which we are especially grateful.

The following, in alphabetical order, have reviewed one or more chapters, correcting our errors of fact or interpretation and helping to ensure they have the appropriate balance and emphasis: Tom Alber (University of California, Berkeley), Tom Blundell (Cambridge University, UK), Stephen Burley (Rockefeller University), Charles Craik (University of California, San Francisco), Ken Dill (University of California, San Francisco), Chris Dobson (Oxford University, UK), Anthony Fink (University of California, Santa Cruz), Robert Fletterick (University of California, San Francisco), Richard Henderson (LMB, Cambridge, UK), Werner Kühlbrandt (MPI, Frankfurt), David Parry (Massey University, New Zealand), Greg Petsko (Brandeis University), and David Trentham (NIMR, London, UK).

The book depends for its accessibility upon its illustrations and we are hugely indebted to Nigel Orme, who, as with the first edition, has converted sketches into lucid figures. Keith Roberts has again advised us on how best graphically to represent chemical and structural phenomena. Jane Richardson (Duke University) has generously produced the Kinemage supplement to this edition and the book relies upon Richardson-type diagrams throughout to render the structures discussed comprehensible. We thank our publishers Garland Publishing, now part of the Taylor and Francis Group, for their support, and in particular Matthew Day for his enthusiastic editing of the complete manuscript. Miranda Robertson, in her inimitable style, has again managed the entire project.

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Basic Structural Principles

Part 1

The Building Blocks

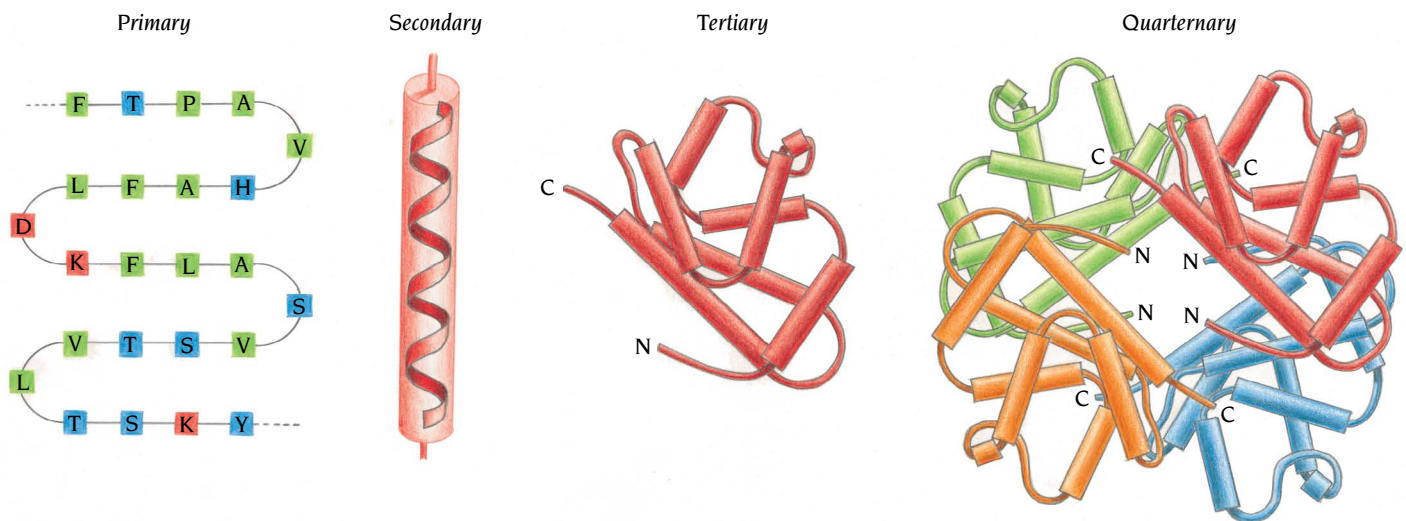
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Recombinant DNA techniques have provided tools for the rapid determination of DNA sequences and, by inference, the amino acid sequences of proteins from structural genes. The number of such sequences is now increasing almost exponentially, but by themselves these sequences tell little more about the biology of the system than a New York City telephone directory tells about the function and marvels of that city.

The proteins we observe in nature have evolved, through selective pressure, to perform specific functions. The functional properties of proteins depend upon their three-dimensional structures. The three-dimensional structure arises because particular sequences of amino acids in polypeptide chains fold to generate, from linear chains, compact domains with specific three-dimensional structures (Figure 1.1). The folded domains can serve as modules for building up large assemblies such as virus particles or muscle fibers, or they can provide specific catalytic or binding sites, as found in enzymes or proteins that carry oxygen or that regulate the function of DNA.

To understand the biological function of proteins we would therefore like to be able to deduce or predict the three-dimensional structure from the amino acid sequence. This we cannot do. In spite of considerable efforts over the past 25 years, this folding problem is still unsolved and remains one of the most basic intellectual challenges in molecular biology.

Figure 1.1 The amino acid sequence of a protein's polypeptide chain is called its **primary** structure. Different regions of the sequence form local regular **secondary** structures, such as alpha (α) helices or beta (β) strands. The **tertiary** structure is formed by packing such structural elements into one or several compact globular units called domains. The final protein may contain several polypeptide chains arranged in a **quaternary** structure. By formation of such tertiary and quaternary structure amino acids far apart in the sequence are brought close together in three dimensions to form a functional region, an **active site**.



Protein folding remains a problem because there are 20 different amino acids that can be combined into many more different proteins than there are atoms in the known universe. In addition there is a vast number of ways in which similar structural domains can be generated in proteins by different amino acid sequences. By contrast, the structure of DNA, made up of only four different nucleotide building blocks that occur in two pairs, is relatively simple, regular, and predictable.

Since the three-dimensional structures of individual proteins cannot be predicted, they must instead be determined experimentally by x-ray crystallography, electron crystallography or nuclear magnetic resonance (NMR) techniques. Over the past 30 years the structures of more than 6000 proteins have been solved, and the sequences of more than 500,000 have been determined. This has generated a body of information from which a set of basic principles of protein structure has emerged. These principles make it easier for us to understand how protein structure is generated, to identify common structural themes, to relate structure to function, and to see fundamental relationships between different proteins. The science of protein structure is at the stage of taxonomy where we can begin to discern patterns and motifs among the relatively small number of proteins whose three-dimensional structure is known.

The first six chapters of this book deal with the basic principles of protein structure as we understand them today, and examples of the different major classes of protein structures are presented. Chapter 7 contains a brief discussion on DNA structures with emphasis on recognition by proteins of specific nucleotide sequences. The remaining chapters illustrate how during evolution different structural solutions have been selected to fulfill particular functions.

Proteins are polypeptide chains

All of the 20 amino acids have in common a central carbon atom (C_α) to which are attached a hydrogen atom, an amino group (NH_2), and a carboxyl group ($COOH$) (Figure 1.2a). What distinguishes one amino acid from another is the side chain attached to the C_α through its fourth valence. There are 20 different side chains specified by the genetic code; others occur, in rare cases, as the products of enzymatic modifications after translation.

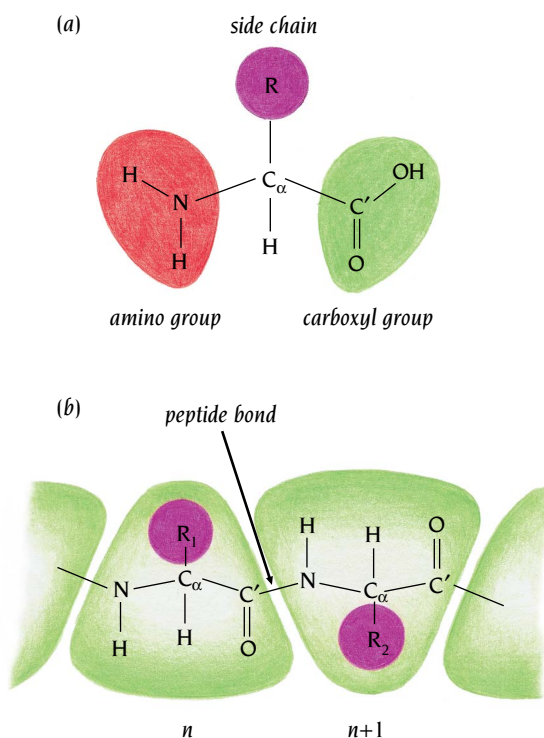
Amino acids are joined end-to-end during protein synthesis by the formation of **peptide bonds** when the carboxyl group of one amino acid condenses with the amino group of the next to eliminate water (Figure 1.2b). This process is repeated as the chain elongates. One consequence is that the amino group of the first amino acid of a polypeptide chain and the carboxyl group of the last amino acid remain intact, and the chain is said to extend from its amino terminus to its carboxy terminus. The formation of a succession of peptide bonds generates a “main chain,” or “backbone,” from which project the various side chains.

The main-chain atoms are a carbon atom C_α to which the side chain is attached, an NH group bound to C_α , and a carbonyl group $C'=O$, where the carbon atom C' is attached to C_α . These units, or residues, are linked into a polypeptide by a peptide bond between the C' atom of one residue and the nitrogen atom of the next (see Figure 1.2b). The basic repeating unit along the main chain from a biochemical or genetic viewpoint is thus $(NH-C_\alpha H-C'=O)$, which is the residue of the common parts of amino acids after peptide bonds have been formed (see Figure 1.2b).

The genetic code specifies 20 different amino acid side chains

The 20 different side chains that occur in proteins are shown in Panel 1.1 (pp. 6–7). Their names are abbreviated with both a three-letter and a one-letter code, which are also given in the panel. The one-letter codes are worth memorizing, as they are widely used in the literature. A mnemonic device for linking the one-letter code to the names of the amino acids is given in Panel 1.1.

Figure 1.2 Proteins are built up by amino acids that are linked by peptide bonds to form a polypeptide chain. (a) Schematic diagram of an amino acid, illustrating the nomenclature used in this book. A central carbon atom (C_α) is attached to an amino group (NH_2), a carboxyl group ($COOH$), a hydrogen atom (H), and a side chain (R). (b) In a polypeptide chain the carboxyl group of amino acid n has formed a peptide bond, C–N, to the amino group of amino acid $n + 1$. One water molecule is eliminated in this process. The repeating units, which are called residues, are divided into main-chain atoms and side chains. The main-chain part, which is identical in all residues, contains a central C_α atom attached to an NH group, a $C'=O$ group, and an H atom. The side chain R, which is different for different residues, is bound to the C_α atom.



The amino acids are usually divided into three different classes defined by the chemical nature of the side chain. The first class comprises those with strictly hydrophobic side chains: Ala (A), Val (V), Leu (L), Ile (I), Phe (F), Pro (P), and Met (M). The four charged residues, Asp (D), Glu (E), Lys (K), and Arg (R), form the second class. The third class comprises those with polar side chains: Ser (S), Thr (T), Cys (C), Asn (N), Gln (Q), His (H), Tyr (Y), and Trp (W). The amino acid glycine (G), which has only a hydrogen atom as a side chain and so is the simplest of the 20 amino acids, has special properties and is usually considered either to form a fourth class or to belong to the first class.

The four groups attached to the C_α atom are chemically different for all the amino acids except glycine, where two H atoms bind to C_α . All amino acids except glycine are thus chiral molecules that can exist in two different forms with different “hands,” L- or D-form (Figure 1.3).

Biological systems depend on specific detailed recognition of molecules that distinguish between chiral forms. The translation machinery for protein synthesis has evolved to utilize only one of the chiral forms of amino acids, the L-form. All amino acids that occur in proteins therefore have the L-form. There is, however, no obvious reason why the L-form was chosen during evolution and not the D-form.

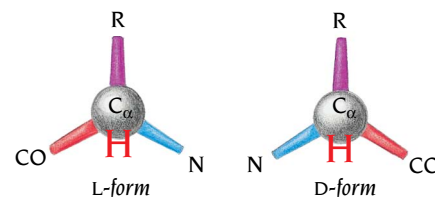
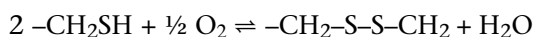


Figure 1.3 The “handedness” of amino acids. Looking down the H– C_α bond from the hydrogen atom, the L-form has CO, R, and N substituents from C_α going in a clockwise direction. There is a mnemonic to remember this; for the L-form the groups read CORN in clockwise direction.

Cysteines can form disulfide bridges

Two cysteine residues in different parts of the polypeptide chain but adjacent in the three-dimensional structure of a protein can be oxidized to form a **disulfide bridge** (Figure 1.4). The disulfide is usually the end product of air oxidation according to the following reaction scheme:



This reaction requires an oxidative environment, and such disulfide bridges are usually not found in intracellular proteins, which spend their lifetime in an essentially reductive environment. Disulfide bridges do, however, occur quite frequently among extracellular proteins that are secreted from cells, and in eucaryotes, formation of these bridges occurs within the lumen of the endoplasmic reticulum, the first compartment of the secretory pathway.

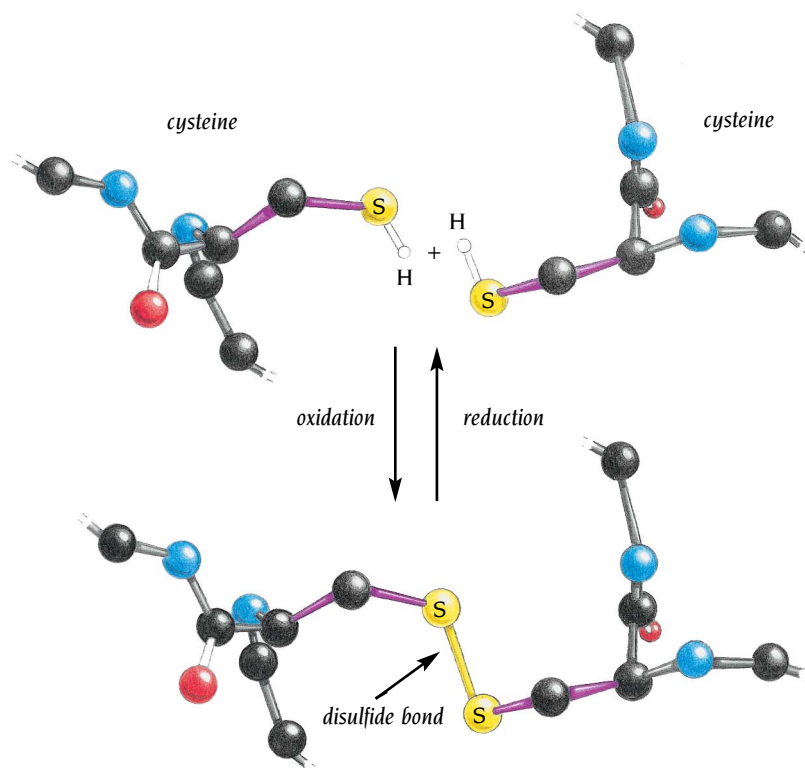
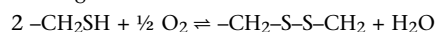
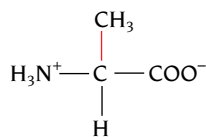
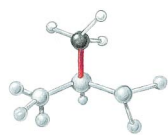


Figure 1.4 The disulfide is usually the end product of air oxidation according to the following schematic reaction scheme:



Disulfide bonds form between the side chains of two cysteine residues. Two SH groups from cysteine residues, which may be in different parts of the amino acid sequence but adjacent in the three-dimensional structure, are oxidized to form one S–S (disulfide) group.

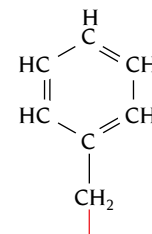
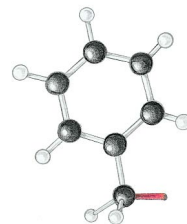
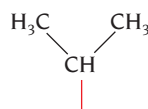
(a) Hydrophobic amino acids



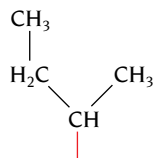
A Ala, Alanine



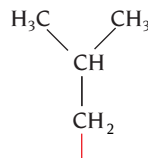
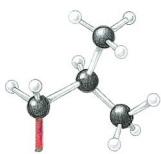
V Val, Valine



F Phe, Phenylalanine

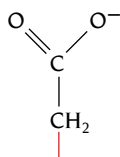
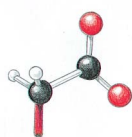


I Ile, Isoleucine

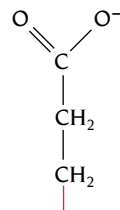
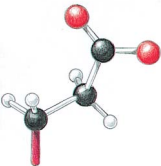


L Leu, Leucine

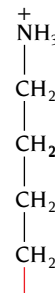
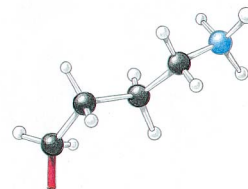
(b) Charged amino acids



D Asp, Aspartic acid

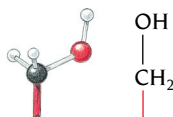


E Glu, Glutamic acid

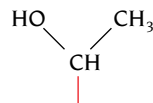


K Lys, Lysine

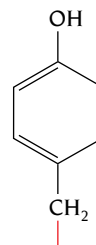
(c) Polar amino acids



S Ser, Serine



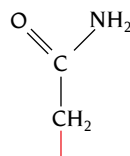
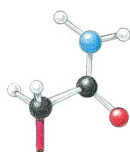
T Thr, Threonine



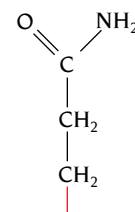
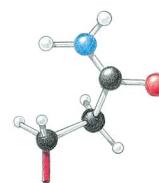
Y Tyr, Tyrosine



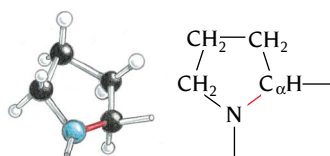
C Cys, Cysteine



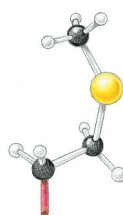
N Asn, Asparagine



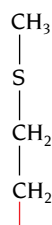
Q Gln, Glutamine



P Pro, Proline



M Met, Methionine



Panel 1.1 The 20 different amino acids that occur in proteins. Only side chains are shown, except for the first amino acid, alanine, where all atoms are shown. The bond from the side chain to C_α is red. A ball-and-stick model, the chemical formula, the full name, and the three-letter and one-letter codes are given for each amino acid.

There are some easy ways of remembering the one-letter code for amino acids. If only one amino acid begins with a certain letter, that letter is used:

C = Cys = Cysteine
H = His = Histidine
I = Ile = Isoleucine
M = Met = Methionine
S = Ser = Serine
V = Val = Valine

If more than one amino acid begins with a certain letter, that letter is assigned to the most commonly occurring amino acid:

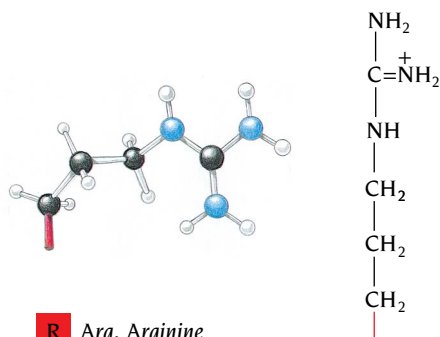
A = Ala = Alanine
G = Gly = Glycine
L = Leu = Leucine
P = Pro = Proline
T = Thr = Threonine

Some of the others are phonetically suggestive:

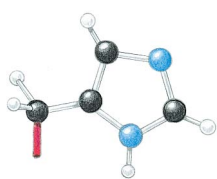
F = Phe = Phenylalanine ("Fenylalanine")
R = Arg = Arginine ("aRginine")
Y = Tyr = Tyrosine ("tYrosine")
W = Trp = Tryptophan (double ring in the molecule)

In other cases a letter close to the initial is used. Amides have letters from the middle of the alphabet. The smaller molecules (D, N, B) are earlier in the alphabet than the larger ones (E, Q, Z).

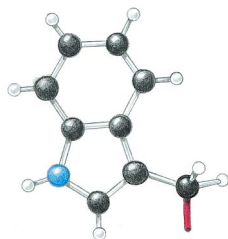
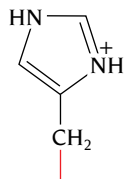
D = Asp = Aspartic acid (near A)
N = Asn = Asparagine (contains N)
B = Asx = Either of D or N
E = Glu = Glutamic acid (near G)
Q = Gln = Glutamine ("Q-tamine")
Z = Glx = Either of E or Q
K = Lys = Lysine (near L)
X = X = Undetermined amino acid



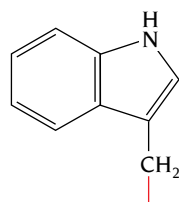
R Arg, Arginine



H His, Histidine



W Trp, Tryptophan



(d) Glycine



G Gly, Glycine

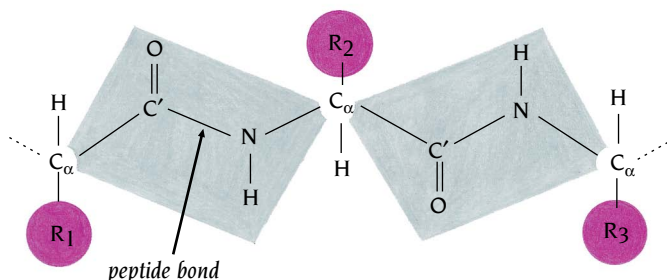


Figure 1.5 Part of a polypeptide chain that is divided into peptide units, represented as blocks in the diagram. Each peptide unit contains the C_α atom and the $C'=O$ group of residue n as well as the NH group and the C_α atom of residue $n + 1$. Each such unit is a planar, rigid group with known bond distances and bond angles. R_1 , R_2 , and R_3 are the side chains attached to the C_α atoms that link the peptide units in the polypeptide chain. The peptide group is planar because the additional electron pair of the $C=O$ bond is delocalized over the peptide group such that rotation around the $C-N$ bond is prevented by an energy barrier.

Disulfide bridges stabilize three-dimensional structure. In some proteins these bridges hold together different polypeptide chains; for example, the A and B chains of insulin are linked by two disulfide bridges between the chains. More frequently intramolecular disulfide bridges stabilize the folding of a single polypeptide chain, making the protein less susceptible to degradation. There are many examples of this, including proteins with short polypeptide chains, such as snake venom toxins and protease inhibitors, that need additional stabilizing factors to produce a stable fold. Much effort is currently spent on introducing extra intramolecular disulfide bridges into enzymes by site-directed mutagenesis in order to make them more thermostable and hence more useful for industrial applications as catalysts, as described in Chapter 17.

Peptide units are building blocks of protein structures

Figure 1.2 shows one way of dividing a polypeptide chain, the biochemist's way. There is, however, a different way to divide the main chain into repeating units that is preferable when we want to describe the structural properties of proteins. For this purpose it is more useful to divide the polypeptide chain into peptide units that go from one C_α atom to the next C_α atom (see Figure 1.5). Each C_α atom, except the first and the last, thus belongs to two such units. The reason for dividing the chain in this way is that all the atoms in such a unit are fixed in a plane with the bond lengths and bond angles very nearly the same in all units in all proteins. Note that the peptide units of the main chain do not involve the different side chains (Figure 1.5). We will use both of these alternative descriptions of polypeptide chains—the biochemical and the structural—and discuss proteins in terms of the sequence of different amino acids and the sequence of planar peptide units.

Since the peptide units are effectively rigid groups that are linked into a chain by covalent bonds at the C_α atoms, the only degrees of freedom they have are rotations around these bonds. Each unit can rotate around two such bonds: the $C_\alpha-C'$ and the $N-C_\alpha$ bonds (Figure 1.6). By convention the angle of rotation around the $N-C_\alpha$ bond is called **phi** (ϕ) and the angle around the $C_\alpha-C'$ bond from the same C_α atom is called **psi** (ψ).

In this way each amino acid residue is associated with two conformational angles ϕ and ψ . Since these are the only degrees of freedom, the conformation of the whole main chain of the polypeptide is completely determined when the ϕ and ψ angles for each amino acid are defined with high accuracy.

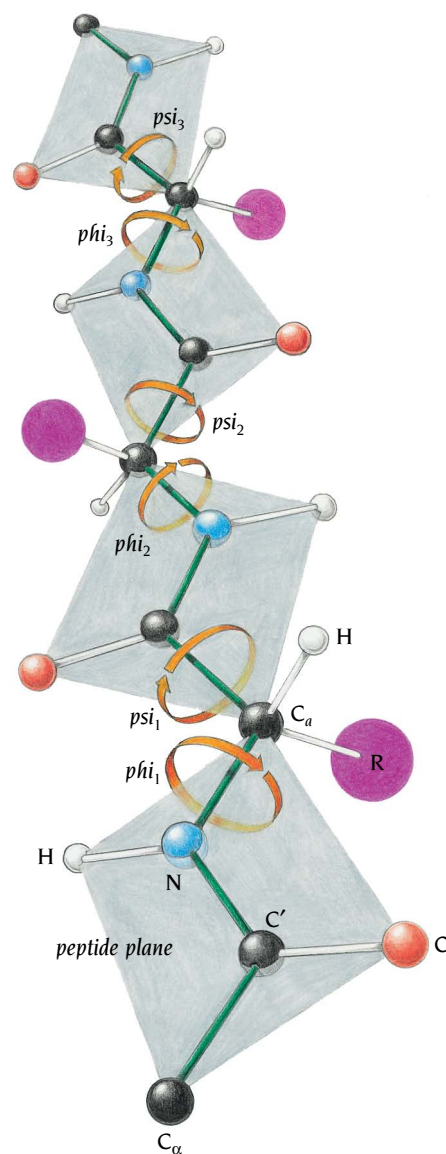


Figure 1.6 Diagram showing a polypeptide chain where the main-chain atoms are represented as rigid peptide units, linked through the C_α atoms. Each unit has two degrees of freedom; it can rotate around two bonds, its $C_\alpha-C'$ bond and its $N-C_\alpha$ bond. The angle of rotation around the $N-C_\alpha$ bond is called phi (ϕ) and that around the $C_\alpha-C'$ bond is called psi (ψ). The conformation of the main-chain atoms is therefore determined by the values of these two angles for each amino acid.

Glycine residues can adopt many different conformations

Most combinations of ϕ and ψ angles for an amino acid are not allowed because of steric collisions between the side chains and main chain. It is reasonably straightforward to calculate those combinations that are allowed. Since the D- and L-forms of the amino acids have their side chain oriented differently with respect to the CO group, they have different allowed ϕ and ψ angles. Proteins built from D-amino acids would thus be expected to have different conformations from those found in nature that are exclusively made of L-amino acids. Since the L- and D-forms of each amino acid are mirror images of one another, would a protein made exclusively of D-form residues produce a structure that is the mirror image of the natural protein? Stephen Kent and his colleagues at the Scripps Institute chemically synthesized both the L- and the D-forms of HIV-1 protease. The D-enzyme proved indeed to be the mirror image of the L-enzyme. Furthermore the D-enzyme and L-enzyme had reciprocal chiral specificity on peptide substrates, the D-enzyme only recognizing and cutting peptides made of D-amino acids. Perhaps the choice of the L-form at the outset of the evolution of life on earth was random and irrevocable.

The angle pairs ϕ and ψ are usually plotted against each other in a diagram called a **Ramachandran plot** after the Indian biophysicist G.N. Ramachandran who first made calculations of sterically allowed regions. Figure 1.7 shows the results of such calculations and also a plot for all amino

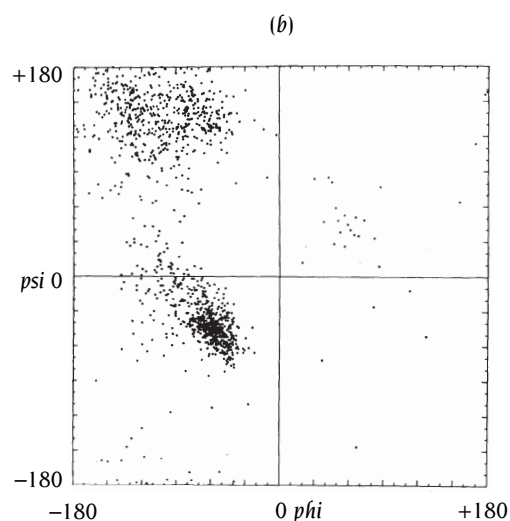
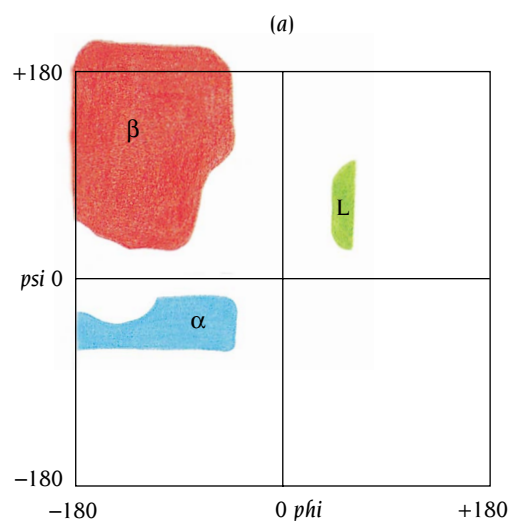
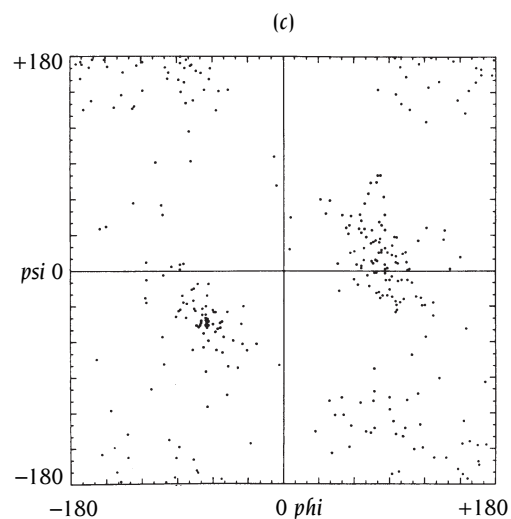


Figure 1.7 Ramachandran plots showing allowed combinations of the conformational angles ϕ and ψ defined in Figure 1.6. Since ϕ (ϕ) and ψ (ψ) refer to rotations of two rigid peptide units around the same C_α atom, most combinations produce steric collisions either between atoms in different peptide groups or between a peptide unit and the side chain attached to C_α . These combinations are therefore not allowed. (a) Colored areas show sterically allowed regions. The areas labeled α , β , and L correspond approximately to conformational angles found for the usual right-handed α helices, β strands, and left-handed α helices, respectively. (b) Observed values for all residue types except glycine. Each point represents ϕ and ψ values for an amino acid residue in a well-refined x-ray structure to high resolution. (c) Observed values for glycine. Notice that the values include combinations of ϕ and ψ that are not allowed for other amino acids. (From J. Richardson, *Adv. Prot. Chem.* 34: 174–175, 1981.)



acids except glycine from a number of accurately determined protein structures. It is apparent that the observed values are clustered in the sterically allowed regions. There is one important exception. Glycine, with only a hydrogen atom as a side chain, can adopt a much wider range of conformations than the other residues, as seen in Figure 1.7c. Glycine thus plays a structurally very important role; it allows unusual main-chain conformations in proteins. This is one of the main reasons why a high proportion of glycine residues are conserved among homologous protein sequences.

Regions in the Ramachandran plot are named after the conformation that results in a peptide if the corresponding ϕ and ψ angles are repeated in successive amino acids along the chain. The major allowed regions in Figure 1.7a are the right-handed α -helical cluster in the lower left quadrant (see Chapter 3); the broad region of extended β strands of both parallel and antiparallel β structures (see Chapter 4) in the upper left quadrant; and the small, sparsely populated left-handed α -helical region in the upper right quadrant. Left-handed α helices are usually found in loop regions or in small single-turn α helices.

Certain side-chain conformations are energetically favorable

Any side chain longer than that of alanine can in principle have several different conformations because of rotations around the bonds between the side-chain carbon atoms. It is a general rule in chemistry that the most energetically favored arrangements for two tetrahedrally coordinated carbon atoms are the “staggered” conformations, in which the substituents of one carbon atom are between those of the other when viewed along the axis of rotation, as illustrated in Figure 1.8. For each such carbon atom in a side chain, there are three possible staggered conformations, which are related by a three step, 120° rotation around the carbon–carbon bond. These three conformations are indistinguishable for alanine, since all three substituents on the side-chain carbon atom C_β are the same.

For valine, however, the three staggered conformations are energetically different because the substituents on C_β are different; two of them are methyl groups and the other is a hydrogen atom (see Figure 1.8b–d). It is energetically most favorable to have the two methyl groups close to the small hydrogen atom bound to C_α , as shown in Figure 1.8b, and therefore this is the conformation most frequently found in proteins. An analysis of accurately determined protein structures has shown that most side chains have one or a few conformations that occur more frequently than the other possible staggered

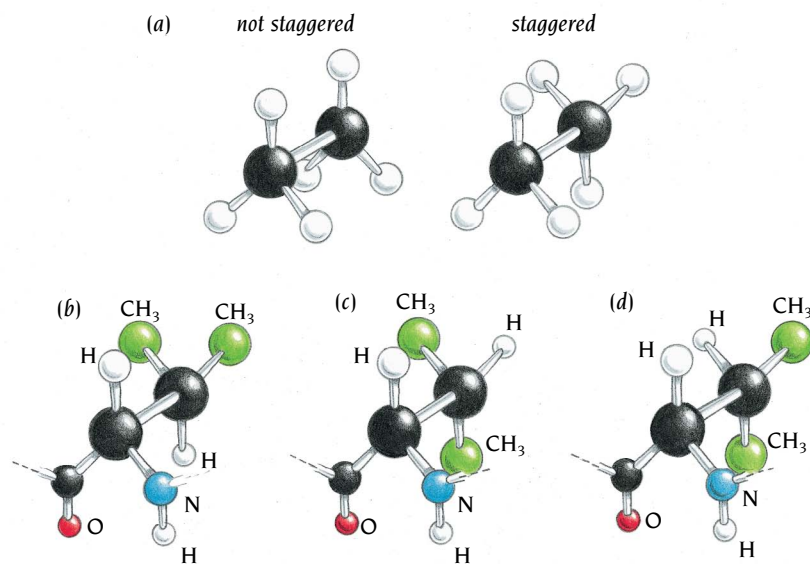
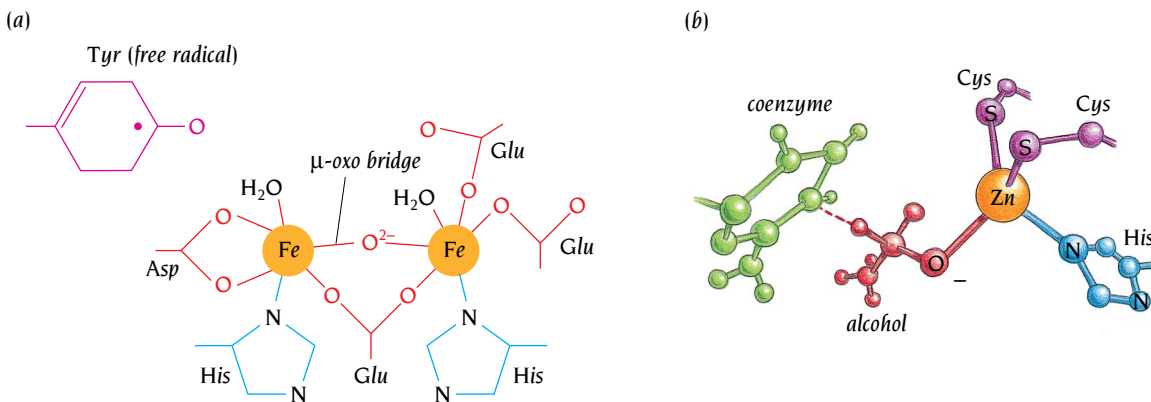


Figure 1.8 The staggered conformations are the most energetically favored conformations of two tetrahedrally coordinated carbon atoms. (a) A view along the C–C bond in ethane (CH_3CH_3) showing how the two carbon atoms can rotate so that their hydrogen atoms are either not staggered (aligned) or staggered. Three indistinguishable staggered conformations are obtained by a rotation of 120° around the C–C bond. (b–d) Similar views as in (a) of valine. The three staggered conformations are different for valine because the three groups attached to C_β are different. The first staggered conformation (b) is less crowded and energetically most favored because the two methyl groups bound to C_β are both close to the small H atom bound to C_α .



conformations. These are called **rotamers**. Today, collections of these favored conformations, or rotamer libraries, are a standard tool in computer programs used for modeling protein structures.

Many proteins contain intrinsic metal atoms

The side chains of the 20 different amino acids listed in Panel 1.1 (pp. 6–7) have very different chemical properties and are utilized for a wide variety of biological functions. However, their chemical versatility is not unlimited, and for some functions metal atoms are more suitable and more efficient. Electron-transfer reactions are an important example. Fortunately the side chains of histidine, cysteine, aspartic acid, and glutamic acid are excellent metal ligands, and a fairly large number of proteins have recruited metal atoms as intrinsic parts of their structures; among the frequently used metals are iron, zinc, magnesium, and calcium. Several metallo proteins are discussed in detail in later chapters and it suffices here to mention briefly a few examples of iron and zinc proteins.

The most conspicuous use of iron in biological systems is in our blood, where the erythrocytes are filled with the oxygen-binding protein hemoglobin. The red color of blood is due to the iron atom bound to the heme group in hemoglobin. Similar heme-bound iron atoms are present in a number of proteins involved in electron-transfer reactions, notably cytochromes. A chemically more sophisticated use of iron is found in an enzyme, ribonucleotide reductase, that catalyzes the conversion of ribonucleotides to deoxyribonucleotides, an important step in the synthesis of the building blocks of DNA.

Ribonucleotide reductase from *Escherichia coli* and mammals contains a di-iron center (Figure 1.9a) that reacts with oxygen and oxidizes a nearby tyrosine side chain, producing a tyrosyl free radical that is essential for the catalysis. The two iron atoms in this iron center are close to each other and bridged by the oxygen atoms of a glutamic acid side chain as well as by an oxygen ion called a μ -oxo bridge. Glutamic acid, aspartic acid, and histidine side chains from the protein, as well as water molecules, complete the coordination sphere of the octahedrally coordinated iron atom.

Zinc is used to stabilize the DNA-binding regions of a class of transcription factors called zinc fingers, which are discussed in Chapter 10. Zinc ions also participate directly in catalytic reactions in many different enzymes by binding substrate molecules and providing a positive charge that influences the electronic arrangement of the substrate and thereby facilitates the catalytic reaction. One such example is found in the enzyme alcohol dehydrogenase, which in yeast produces alcohol during fermentation and in our livers detoxifies the alcohol we have consumed by oxidizing it. The enzyme provides a scaffold containing three zinc ligands—one histidine and two cysteine side chains—which sequester a zinc atom so that it binds alcohol as a fourth ligand in a tetrahedral coordination (Figure 1.9b).

Figure 1.9 Examples of functionally important intrinsic metal atoms in proteins. (a) The di-iron center of the enzyme ribonucleotide reductase. Two iron atoms form a redox center that produces a free radical in a nearby tyrosine side chain. The iron atoms are bridged by a glutamic acid residue and a negatively charged oxygen atom called a μ -oxo bridge. The coordination of the iron atoms is completed by histidine, aspartic acid, and glutamic acid side chains as well as water molecules. (b) The catalytically active zinc atom in the enzyme alcohol dehydrogenase. The zinc atom is coordinated to the protein by one histidine and two cysteine side chains. During catalysis zinc binds an alcohol molecule in a suitable position for hydride transfer to the coenzyme moiety, a nicotinamide. [(a) Adapted from P. Nordlund et al., *Nature* 345: 593–598, 1990.]

Conclusion

All protein molecules are polymers built up from 20 different amino acids linked end-to-end by peptide bonds. The function of every protein molecule depends on its three-dimensional structure, which in turn is determined by its amino acid sequence, which in turn is determined by the nucleotide sequence of the structural gene.

Each amino acid has atoms in common, and these form the main chain of the protein. The remaining atoms form side chains that can be hydrophobic, polar, or charged.

The conformation of the whole main chain of a protein is determined by two conformational angles, phi (ϕ) and psi (ψ), for each amino acid. Only certain combinations of these angles are allowed because of steric hindrance between main-chain atoms and side-chain atoms, except for glycine.

Certain side-chain conformations are energetically more favorable than others. Computer programs used to model protein structures contain rotamer libraries of such favored conformations.

Many proteins contain intrinsic metal atoms that are functionally important. The most frequently used metals are iron, zinc, magnesium, and calcium. These metal atoms are mainly bound to the protein through the side chains of cysteine, histidine, aspartic acid, and glutamic acid residues.

Selected readings

Alberts, B., et al. *Molecular Biology of the Cell*, 3rd ed. New York: Garland, 1994.

Creighton, T.E. *Proteins: Structures and Molecular Properties*, 2nd ed. New York: Freeman, 1993.

Fletterick, R.J., Schroer, T., Matela, R.J. *Molecular Structure: Macromolecules in Three Dimensions*. Oxford, UK: Blackwell Scientific, 1985.

Judson, H.F. *The Eighth Day of Creation: Makers of the Revolution in Biology*. New York: Simon & Schuster, 1979.

Karlin, K.D. Metalloenzymes, structural motifs, and inorganic models. *Science* 261: 701–708, 1993.

Lesk, A. *Protein Architecture: A Practical Approach*. Oxford, UK: Oxford University Press, 1991.

Mathews, C.K., van Holde, K.E. *Biochemistry*. Menlo Park, CA: Benjamin/Cummings, 1990.

Perutz, M. *Protein Structure: New Approaches to Disease and Therapy*. New York: Freeman, 1992.

Petsko, G.A. On the other hand... *Science* 256: 1403–1404, 1992.

Ramachandran, G.N., Sasisekharan, V. Conformation of polypeptides and proteins. *Adv. Prot. Chem.* 28: 283–437, 1968.

Richardson, J.S., Richardson, D.C. Principles and patterns of protein conformation. In *Prediction of Protein Structure and the Principles of Protein Conformation* (ed. Fasman, G.D.), pp. 1–98. New York: Plenum, 1989.

Schulz, G.E., Schirmer, R.H. *Principles of Protein Structure*. New York: Springer, 1979.

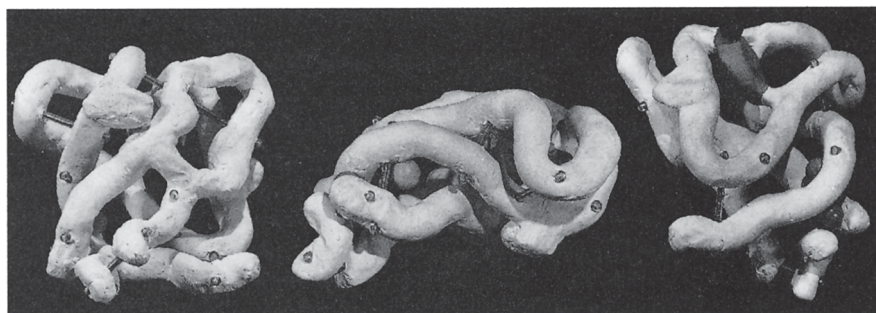
Spiro, T.S. *Zinc Enzymes*. New York: Wiley, 1983.

Stryer, L. *Biochemistry*, 4th ed. New York: Freeman, 1995.

X-ray structural studies have played a major role in transforming chemistry from a descriptive science at the beginning of the twentieth century to one in which the properties of novel compounds can be predicted on theoretical grounds. When W.L. Bragg solved the very first crystal structure, that of rock salt, NaCl, the results completely changed prevalent concepts of bonding forces in ionic compounds.

The first x-ray crystallographic structural results on a globular protein molecule were reported for myoglobin (Figure 2.1) in 1958, and came as a shock to those who had hoped for simple, general principles of protein structure and function analogous to the simple and beautiful double-stranded DNA structure that had been determined five years before by James Watson and Francis Crick. John Kendrew at the Medical Research Council Laboratory of Molecular Biology, in Cambridge, UK, who determined the myoglobin structure to low resolution in 1958, expressed his disappointment about the complexity of the structure in the following words: "Perhaps the most remarkable features of the molecule are its complexity and its lack of symmetry. The arrangement seems to be almost totally lacking in the kind of regularities which one instinctively anticipates, and it is more complicated than has been predicted by any theory of protein structure."

In retrospect it is easy to see that such structural irregularity is actually required for proteins to fulfill their diverse functions. Information storage and transfer from DNA is essentially linear, and DNA molecules of very different information content can therefore have essentially the same gross structure. In contrast, proteins must recognize many thousands of different molecules in the cell by detailed three-dimensional interactions, which



10 Å

Figure 2.1 Kendrew's model of the low-resolution structure of myoglobin shown in three different views. The sausage-shaped regions represent α helices, which are arranged in a seemingly irregular manner to form a compact globular molecule. (Courtesy of J.C. Kendrew.)

require diverse and irregular structures of the protein molecules. In spite of these requirements, there are regular features in protein structures, the most important of which is their **secondary structure**.

The interior of proteins is hydrophobic

When high-resolution studies of myoglobin became available, Kendrew noticed that the amino acids in the interior of the protein had almost exclusively hydrophobic side chains. This was one of the first important general principles to emerge from studies of protein structure. The main driving force for folding water-soluble globular protein molecules is to pack hydrophobic side chains into the interior of the molecule, thus creating a **hydrophobic core** and a hydrophilic surface.

The hydrophobic core is surprisingly densely packed with the side chains in the interior of the protein. Given the constraints of the different shapes of the hydrophobic side chains and considering that their positions must be compatible with the regular secondary structure in the interior of the protein, fitting these shapes into a densely packed core is like solving a three-dimensional jigsaw puzzle. In the few cases where there is a hole in the interior, the space is usually occupied by one or more water molecules that hydrogen-bond to internal polar groups. Such firmly bound internal water molecules can be regarded as integral parts of the protein structure.

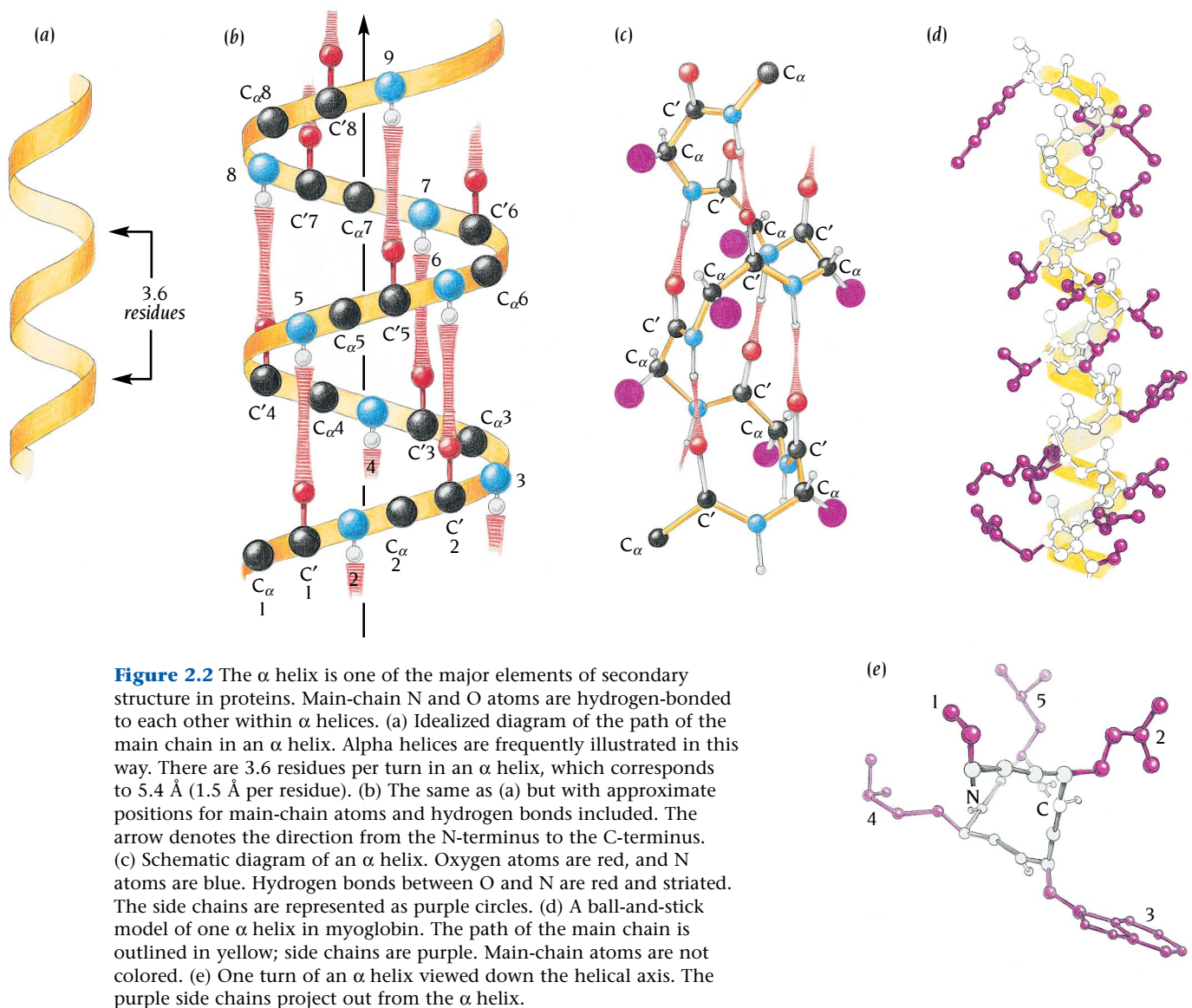
There is a major problem, however, with creating such a hydrophobic core from a protein chain. To bring the side chains into the core, the main chain must also fold into the interior. The main chain is highly polar and therefore hydrophilic, with one hydrogen bond donor, NH, and one hydrogen bond acceptor, C=O, for each peptide unit. In a hydrophobic environment, these main-chain polar groups must be neutralized by the formation of hydrogen bonds. This problem is solved in a very elegant way by the formation of regular secondary structure within the interior of the protein molecule. Such secondary structure is usually one of two types: **alpha helices** or **beta sheets**. Both types are characterized by hydrogen-bonding between the main-chain NH and C=O groups, and they are formed when a number of consecutive residues have the same phi (ϕ), psi (ψ) angles.

The secondary structure elements, formed in this way and held together by the hydrophobic core, provide a rigid and stable framework. They exhibit relatively little flexibility with respect to each other, and they are the best-defined parts of protein structures determined by both x-ray and NMR techniques. Functional groups of the protein are attached to this framework, either directly by their side chains or, more frequently, in loop regions that connect sequentially adjacent secondary structure elements. We will now have a closer look at these structural elements.

The alpha (α) helix is an important element of secondary structure

The α helix is the classic element of protein structure. It was first described in 1951 by Linus Pauling working at the California Institute of Technology. He predicted that it was a structure which would be stable and energetically favorable in proteins. He made this remarkable prediction on the basis of accurate geometrical parameters that he had derived for the peptide unit from the results of crystallographic analyses of the structures of a range of small molecules. This prediction almost immediately received strong experimental support from diffraction patterns obtained by Max Perutz in Cambridge, UK, from hemoglobin crystals and keratin fibers. It was completely verified from John Kendrew's high-resolution structure of myoglobin, where all secondary structure is helical.

Alpha helices in proteins are found when a stretch of consecutive residues all have the ϕ , ψ angle pair approximately -60° and -50° , corresponding to the allowed region in the bottom left quadrant of the



Ramachandran plot (see Figure 1.7a). The α helix has 3.6 residues per turn with hydrogen bonds between $C'=O$ of residue n and NH of residue $n + 4$ (Figure 2.2). Thus all NH and $C'O$ groups are joined with hydrogen bonds except the first NH groups and the last $C'O$ groups at the ends of the α helix. As a consequence, the ends of α helices are polar and are almost always at the surface of protein molecules.

Variations on the α helix in which the chain is either more loosely or more tightly coiled, with hydrogen bonds to residues $n + 5$ or $n + 3$ instead of $n + 4$ are called the π helix and 3_{10} helix, respectively. The 3_{10} helix has 3 residues per turn and contains 10 atoms between the hydrogen bond donor and acceptor, hence its name. Both the π helix and the 3_{10} helix occur rarely and usually only at the ends of α helices or as single-turn helices. They are not energetically favorable, since the backbone atoms are too tightly packed in the 3_{10} helix and so loosely packed in the π helix that there is a hole through the middle. Only in the α helix are the backbone atoms properly packed to provide a stable structure.

In globular proteins α helices vary considerably in length, ranging from four or five amino acids to over forty residues. The average length is around ten residues, corresponding to three turns. The rise per residue of an α helix is 1.5 Å along the helical axis, which corresponds to about 15 Å from one end to the other of an average α helix.

An α helix can in theory be either right-handed or left-handed depending on the screw direction of the chain. A left-handed α helix is not, however, allowed for L-amino acids due to the close approach of the side chains and the C'O group. Thus the α helix that is observed in proteins is almost always right-handed. Short regions of left-handed α helices (3–5 residues) occur only occasionally.

The α helix has a dipole moment

All the hydrogen bonds in an α helix point in the same direction because the peptide units are aligned in the same orientation along the helical axis. Since a peptide unit has a dipole moment arising from the different polarity of NH and C'O groups, these dipole moments are also aligned along the helical axis (Figure 2.3). The overall effect is a significant net dipole for the α helix that gives a partial positive charge at the amino end and a partial negative charge at the carboxy end of the α helix. The magnitude of this dipole moment corresponds to about 0.5–0.7 unit charge at each end of the helix. These charges would be expected to attract ligands of opposite charge and negatively charged ligands, especially when they contain phosphate groups and frequently bind at the N-termini of α helices. In contrast, positively charged ligands rarely bind at the C-terminus. This may be because, in addition to the dipole effect, the N-terminus of an α helix has free NH groups with favorable geometry to position phosphate groups by specific hydrogen bonds (see Figure 2.3). Such ligand-binding occurs frequently in proteins; it provides examples of specific binding through main-chain conformation in which side chains are not involved.

Some amino acids are preferred in α helices

The amino acid side chains project out from the α helix (see Figure 2.2e) and do not interfere with it, except for proline. The last atom of the proline side

Figure 2.3 Negatively charged groups such as phosphate ions frequently bind to the amino ends of α helices. The dipole moment of an α helix as well as the possibility of hydrogen-bonding to free NH groups at the end of the helix favors such binding. (a) The dipole of a peptide unit. Values in boxes give the approximate fractional charges of the atoms of the peptide unit. (b) The dipoles of peptide units are aligned along the α -helical axis, which creates an overall dipole moment of the α helix, positive at the amino end and negative at the carboxy end. (c) A phosphate group hydrogen-bonded to the NH end of an α helix. Nitrogen atoms are blue; oxygen atoms are red; main-chain carbon atoms are black; and phosphorus is green.

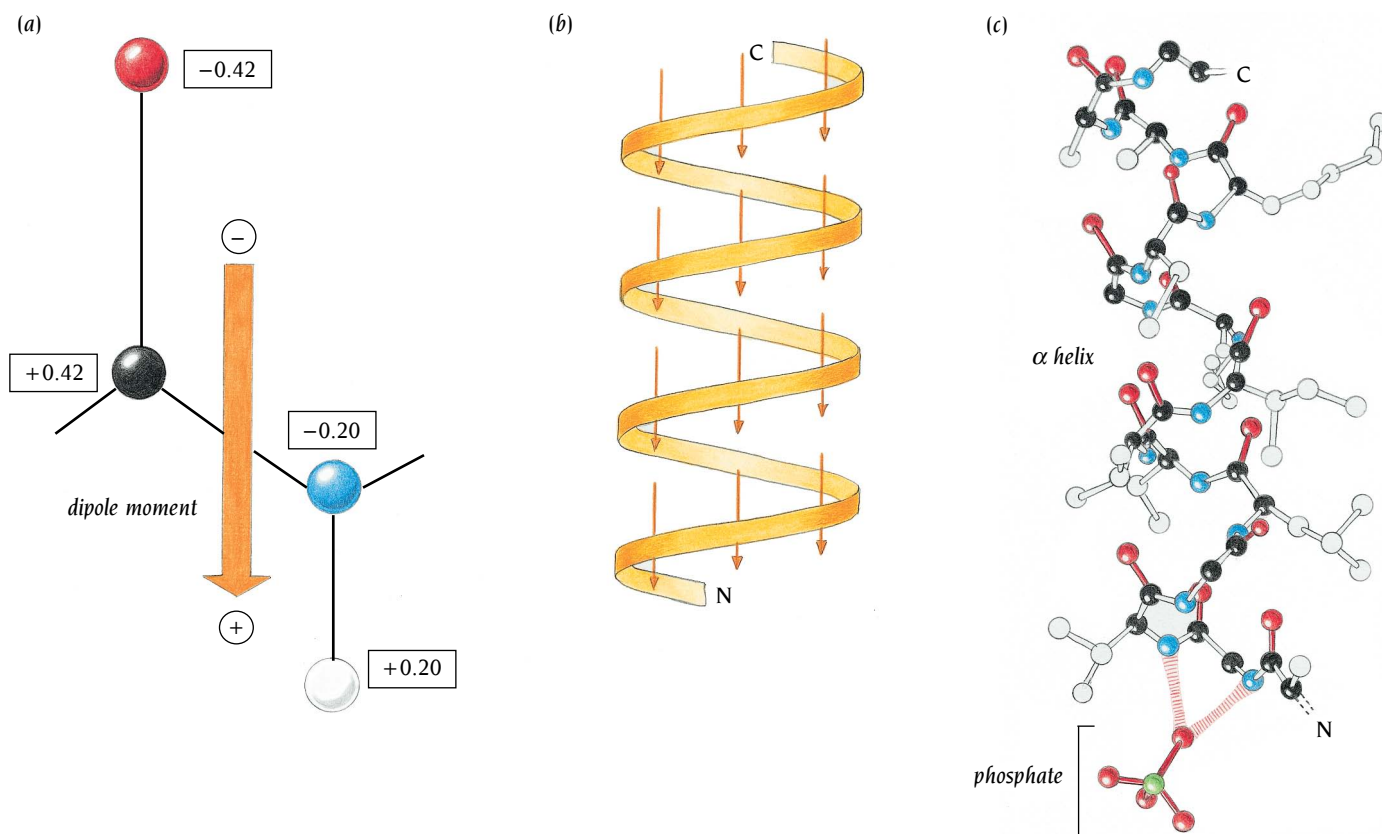


Table 2.1 Amino acid sequences of three α helices

1. - **Leu** - **Ser** - **Phe** - **Ala** - **Ala** - **Ala** - **Met** - **Asn** - **Gly** - **Leu** - **Ala** -
2. - **Ile** - **Asn** - **Glu** - **Gly** - **Phe** - **Asp** - **Leu** - **Leu** - **Arg** - **Ser** - **Gly** -
3. - **Lys** - **Glu** - **Asp** - **Ala** - **Lys** - **Gly** - **Lys** - **Ser** - **Glu** - **Glu** - **Glu** -

The first sequence is from the enzyme citrate synthase, residues 260–270, which form a buried helix; the second sequence is from the enzyme alcohol dehydrogenase, residues 355–365, which form a partially exposed helix; and the third sequence is from troponin-C, residues 87–97, which form a completely exposed helix. Charged residues are colored red, polar residues are blue, and hydrophobic residues are green.

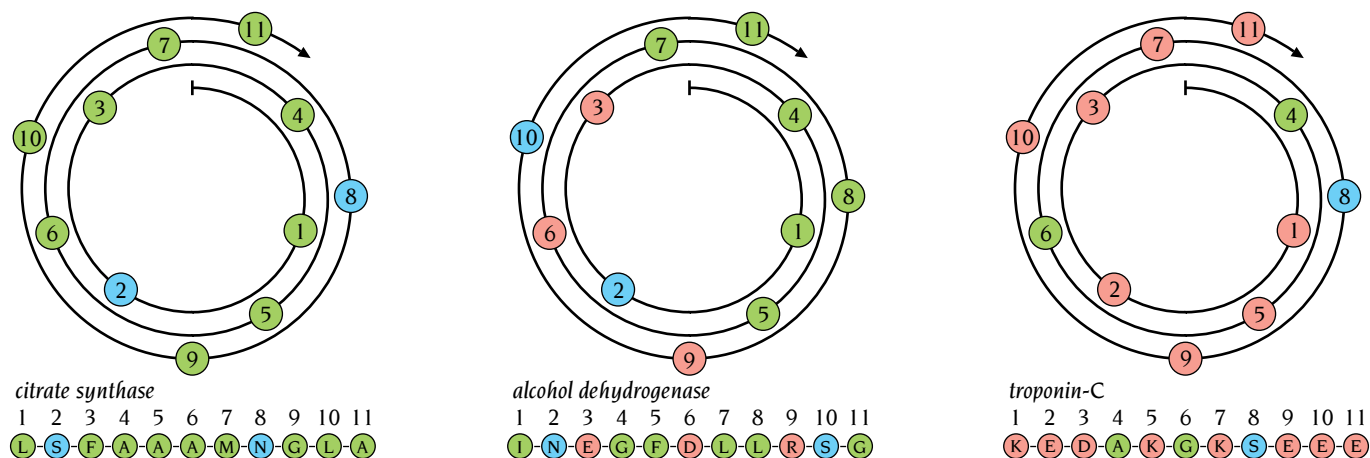
chain is bonded to the main-chain N atom, which forms a ring structure, $C_{\alpha}-CH_2-CH_2-CH_2-N$ (see Panel 1.1, p. 7). This prevents the N atom from participating in hydrogen-bonding and also provides some steric hindrance to the α -helical conformation. Proline fits very well in the first turn of an α helix, but it usually produces a significant bend if it is anywhere else in the helix. Such bends occur in many α helices, not just in those few that contain a proline in the middle. Therefore, although we can predict that a proline residue may cause a bend in an α helix, it does not follow that all bends result from the presence of proline.

Different side chains have been found to have weak but definite preferences either for or against being in α helices. Thus Ala (A), Glu (E), Leu (L), and Met (M) are good α -helix formers, while Pro (P), Gly (G), Tyr (Y), and Ser (S) are very poor. Such preferences were central to all early attempts to predict secondary structure from amino acid sequence, but they are not strong enough to give accurate predictions.

The most common location for an α helix in a protein structure is along the outside of the protein, with one side of the helix facing the solution and the other side facing the hydrophobic interior of the protein. Therefore, with 3.6 residues per turn, there is a tendency for side chains to change from hydrophobic to hydrophilic with a periodicity of three to four residues. Although this trend can sometimes be seen in the amino acid sequence, it is not strong enough for reliable structural prediction by itself, because residues that face the solution can be hydrophobic and, furthermore, α helices can be either completely buried within the protein or completely exposed. Table 2.1 shows examples of the amino acid sequences of a totally buried, a partially buried, and a completely exposed α helix.

A convenient way to illustrate the amino acid sequences in helices is the **helical wheel** or spiral. Since one turn in an α helix is 3.6 residues long, each residue can be plotted every $360/3.6 = 100^\circ$ around a circle or a spiral, as shown in Figure 2.4. Such a plot shows the projection of the position of the

Figure 2.4 The helical wheel or spiral. Amino acid residues are plotted every 100° around the spiral, following the sequences given in Table 2.1. The following color code is used: green is an amino acid with a hydrophobic side chain, blue is a polar side chain, and red is a charged side chain. The first helix is all hydrophobic, the second is polar on one side and hydrophobic on the other side, and the third helix is all polar.



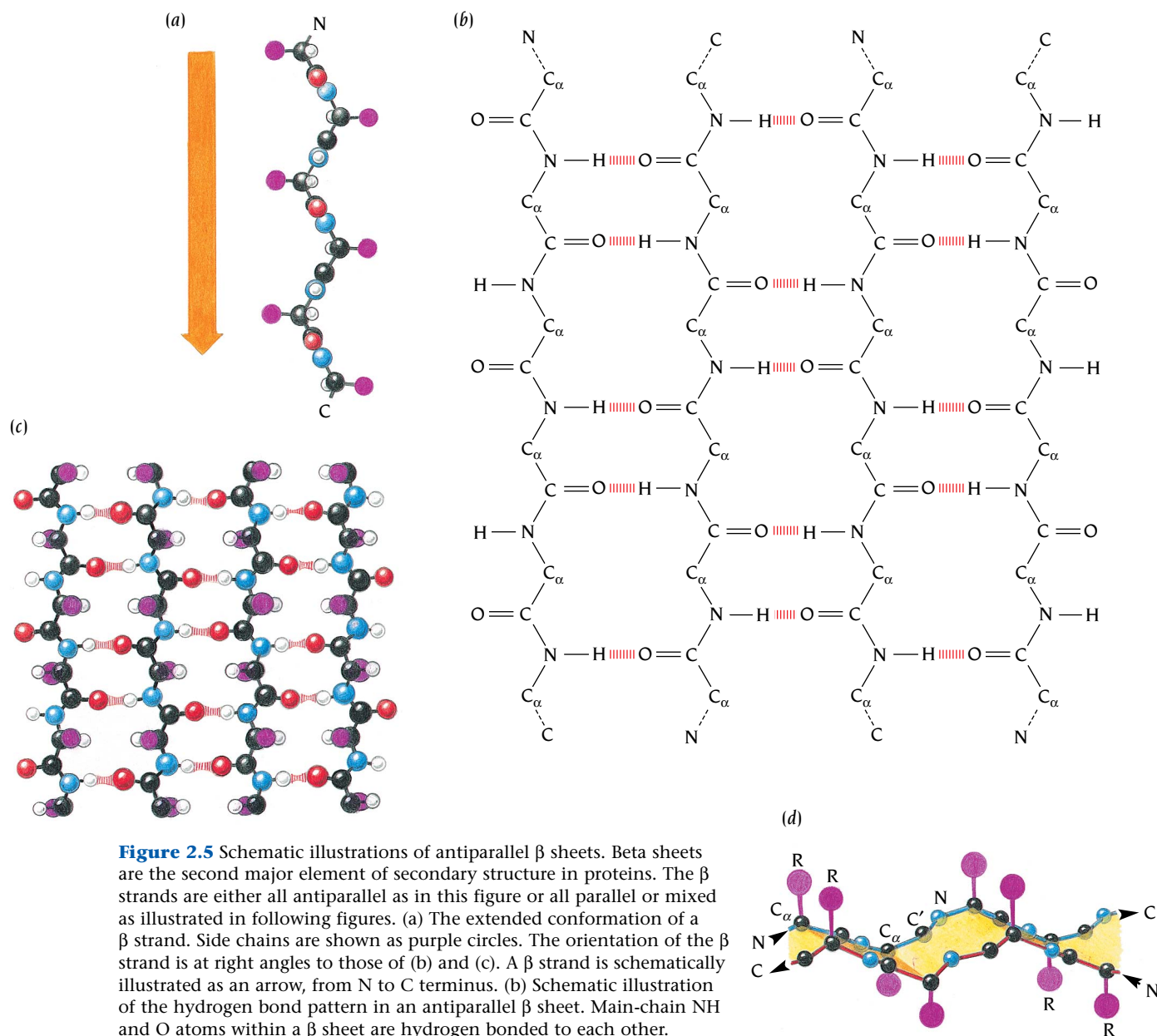
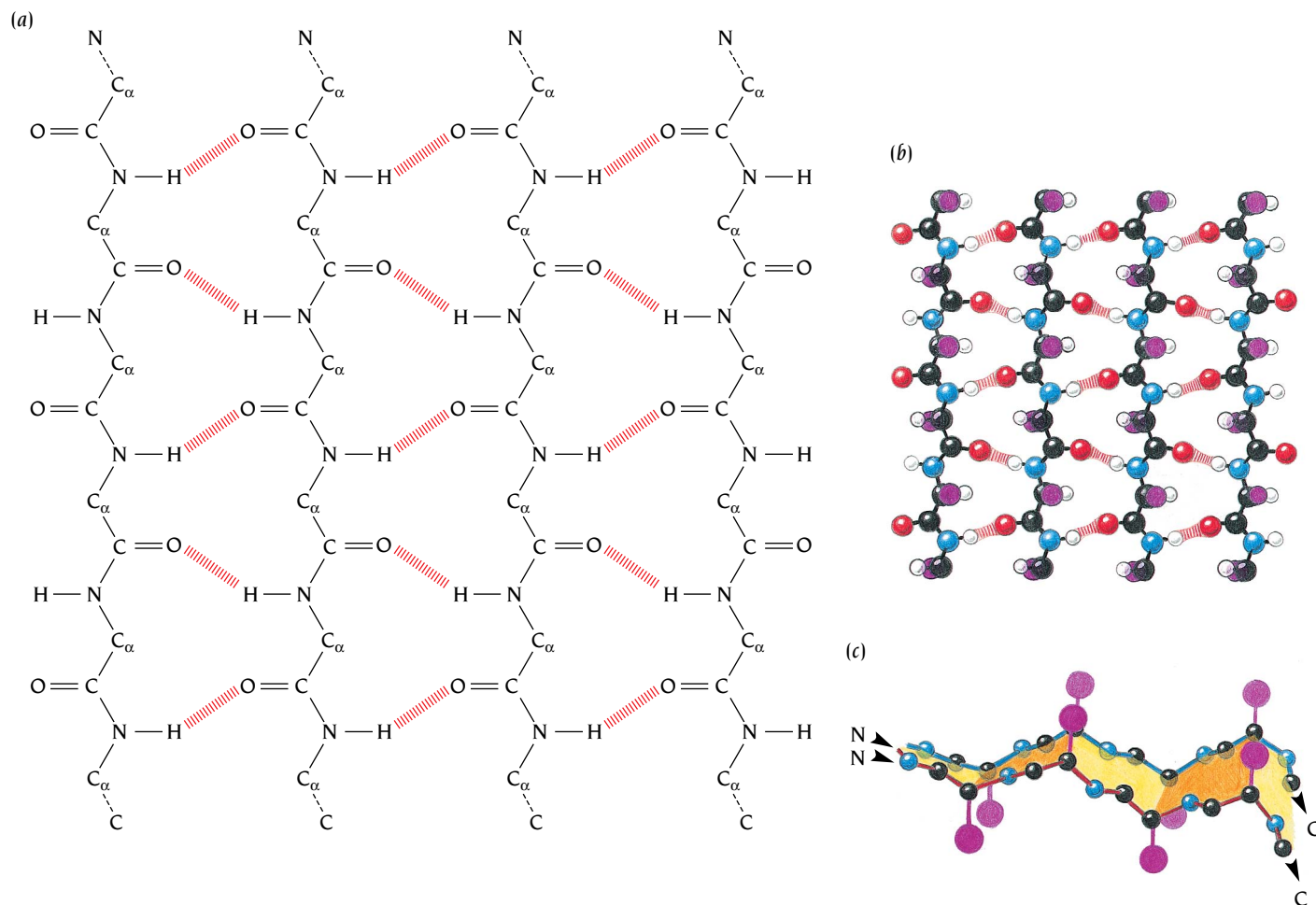


Figure 2.5 Schematic illustrations of antiparallel β sheets. Beta sheets are the second major element of secondary structure in proteins. The β strands are either all antiparallel as in this figure or all parallel or mixed as illustrated in following figures. (a) The extended conformation of a β strand. Side chains are shown as purple circles. The orientation of the β strand is at right angles to those of (b) and (c). A β strand is schematically illustrated as an arrow, from N to C terminus. (b) Schematic illustration of the hydrogen bond pattern in an antiparallel β sheet. Main-chain NH and O atoms within a β sheet are hydrogen bonded to each other. (c) A ball-and-stick version of (b). Oxygen atoms are red; nitrogen atoms are blue. The hydrogen atom in N-H...O is white. The carbon atom in the main chain, C_α , is black. Side chains are illustrated by one purple atom. The orientation of the β strands is different from that in (a). (d) Illustration of the pleat of a β sheet. Two antiparallel β strands are viewed from the side of the β sheet. Note that the directions of the side chains, R (purple), follow the pleat, which is emphasized in yellow.

residues onto a plane perpendicular to the helical axis. Residues on one side of the helix are plotted on one side of the spiral. The three helices whose sequences are given in Table 2.1 are plotted in this way in Figure 2.4, using a color code for hydrophobic, polar, and charged residues. It is immediately obvious that one side of the helix from alcohol dehydrogenase is hydrophilic and the other side hydrophobic.

Alpha helices that cross membranes are in a hydrophobic environment. Therefore, most of their side chains are hydrophobic. Long regions of hydrophobic residues in the amino acid sequence of a protein that is membrane-bound can therefore be predicted with a high degree of confidence to be transmembrane helices, as will be discussed in Chapter 12.



Beta (β) sheets usually have their β strands either parallel or antiparallel

The second major structural element found in globular proteins is the β sheet. This structure is built up from a combination of several regions of the polypeptide chain, in contrast to the α helix, which is built up from one continuous region. These regions, β strands, are usually from 5 to 10 residues long and are in an almost fully extended conformation with ϕ , ψ angles within the broad structurally allowed region in the upper left quadrant of the Ramachandran plot (see Figure 1.7). These β strands are aligned adjacent to each other (see Figures 2.5 and 2.6) such that hydrogen bonds can form between C'=O groups of one β strand and NH groups on an adjacent β strand and vice versa. The β sheets that are formed from several such β strands are "**pleated**" with C α atoms successively a little above and below the plane of the β sheet. The side chains follow this pattern such that within a β strand they also point alternately above and below the β sheet.

Beta strands can interact in two ways to form a pleated sheet. Either the amino acids in the aligned β strands can all run in the same biochemical direction, amino terminal to carboxy terminal, in which case the sheet is described as **parallel**, or the amino acids in successive strands can have alternating directions, amino terminal to carboxy terminal followed by carboxy terminal to amino terminal, followed by amino terminal to carboxy terminal, and so on, in which case the sheet is called **antiparallel**. Each of the two forms has a distinctive pattern of hydrogen-bonding. The antiparallel β sheet (Figure 2.5) has narrowly spaced hydrogen bond pairs that alternate with widely spaced pairs. Parallel β sheets (Figure 2.6) have evenly spaced hydrogen bonds that bridge the β strands at an angle. Within both types of β sheets all possible main-chain hydrogen bonds are formed, except for the two flanking strands of the β sheet that have only one neighboring β strand.

Figure 2.6 Parallel β sheet. (a) Schematic diagram showing the hydrogen bond pattern in a parallel β sheet. (b) Ball-and-stick version of (a). The same color scheme is used as in Figure 2.5c. (c) Schematic diagram illustrating the pleat of a parallel β sheet.

(a)

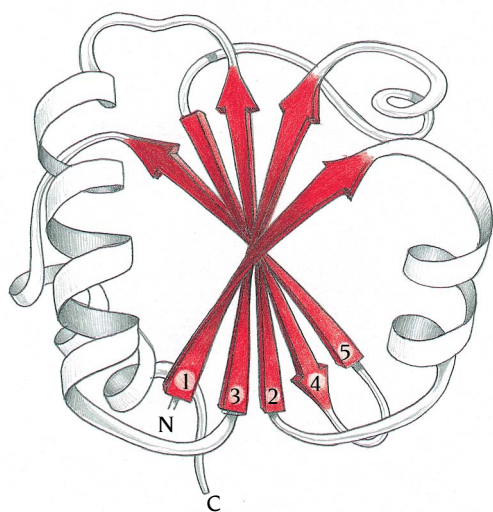
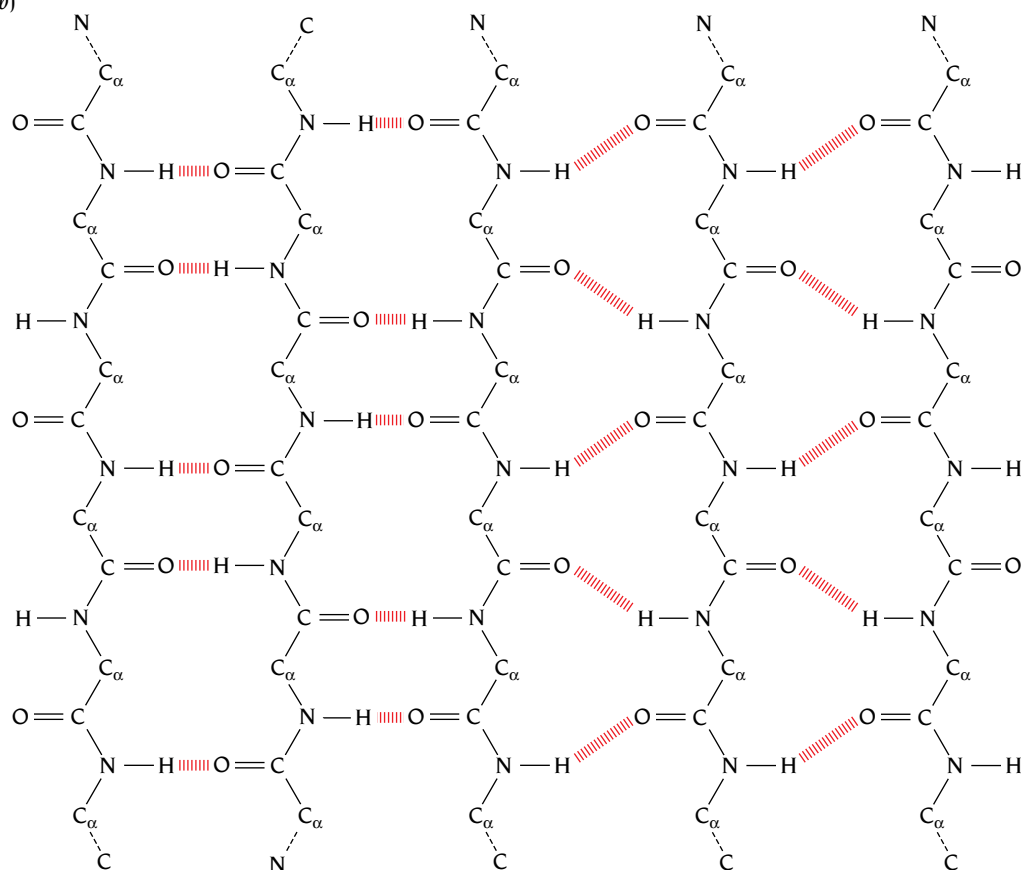


Figure 2.7 (a) Illustration of the twist of β sheets. Beta strands are drawn as arrows from the amino end to the carboxy end of the β strand in this schematic drawing of the protein thioredoxin from *E. coli*, the structure of which was determined in the laboratory of Carl Branden, Uppsala, Sweden, to 2.8 Å resolution. The mixed β sheet is viewed from one of its ends. (b) The hydrogen bonds between the β strands in the mixed β sheet of the same protein. [(a) Adapted from B. Furugren.]

(b)



Beta strands can also combine into mixed β sheets with some β strand pairs parallel and some antiparallel. There is a strong bias against mixed β sheets; only about 20% of the strands inside the β sheets of known protein structures have parallel bonding on one side and antiparallel bonding on the other. Figure 2.7 illustrates how the hydrogen bonds between the β strands are arranged in a mixed β sheet.

As they occur in known protein structures, almost all β sheets—parallel, antiparallel, and mixed—have twisted strands. This twist always has the same handedness as that shown in Figure 2.7, which is defined as a right-handed twist.

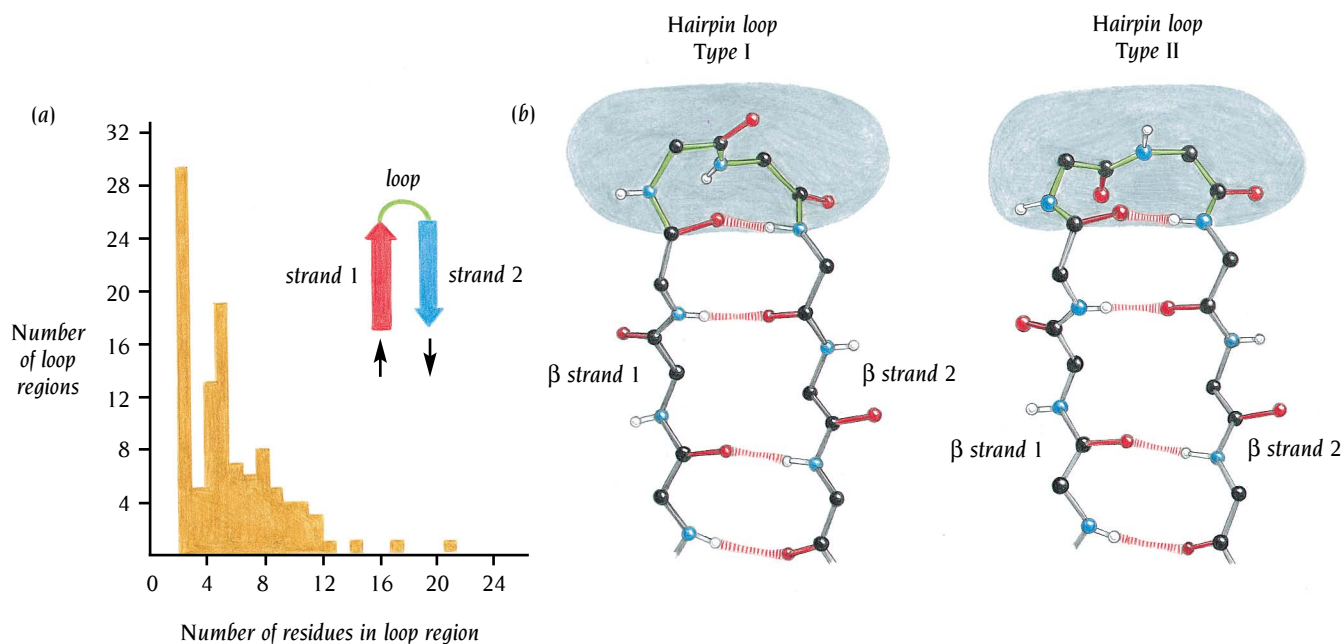
Loop regions are at the surface of protein molecules

Most protein structures are built up from combinations of secondary structure elements, α helices and β strands, which are connected by **loop regions** of various lengths and irregular shape. A combination of secondary structure elements forms the stable hydrophobic core of the molecule. The loop regions are at the surface of the molecule. The main-chain C=O and NH groups of these loop regions, which in general do not form hydrogen bonds to each other, are exposed to the solvent and can form hydrogen bonds to water molecules.

Loop regions exposed to solvent are rich in charged and polar hydrophilic residues. This has been used in several prediction schemes, and it has proved possible to predict loop regions from an amino acid sequence with a higher degree of confidence than α helices or β strands, which is ironic since the loops have irregular structures.

When homologous amino acid sequences from different species are compared, it is found that insertions and deletions of a few residues occur almost exclusively in the loop regions. During evolution, cores are much more stable than loops. Intron positions are also often found at sites in structural genes that correspond to loop regions in the protein structure. Since proteins that exhibit sequence homology in general have similar core structures, it is apparent that the specific arrangement of secondary structure elements in the core is rather insensitive to the lengths of the loop regions. In addition to their function as connecting units between secondary structure elements, loop regions frequently participate in forming binding sites and enzyme active sites. Thus antigen-binding sites in antibodies are built up from six loop regions, which vary both in length and in amino acid sequence between different antibodies. Modeling antigen-binding sites from a known antibody sequence (discussed in Chapter 17) is thus essentially a problem of modeling the three-dimensional structures of loop regions since the core structures of all antibodies are very similar. Such model building has been facilitated by the recent findings that loop regions have preferred structures. Surveys of known three-dimensional structures of loops have shown that they fall into a rather limited set of structures and are not a random collection of possible structures (Figure 2.8). Loop regions that connect two adjacent antiparallel β strands are called **hairpin loops**. Short hairpin loops are usually called **reverse turns** or simply turns. Figure 2.8b shows two of the most frequently

Figure 2.8 Adjacent antiparallel β strands are joined by hairpin loops. Such loops are frequently short and do not have regular secondary structure. Nevertheless, many loop regions in different proteins have similar structures. (a) Histogram showing the frequency of hairpin loops of different lengths in 62 different proteins. (b) The two most frequently occurring two-residue hairpin loops; Type I turn to the left and Type II turn to the right. Bonds within the hairpin loop are green. [(a) Adapted from B.L. Sibanda and J.M. Thornton, *Nature* 316: 170–174, 1985.]



occurring turns; the Type I turn and the Type II turn. The Type II turn usually has a glycine residue as the second of the two residues in the turn.

Long loop regions are often flexible and can frequently adopt several different conformations, making them “invisible” in x-ray structure determinations and undetermined in NMR studies. Such loops are frequently involved in the function of the protein and can switch from an “open” conformation, which allows access to the active site, to a “closed” conformation, which shields reactive groups in the active site from water.

Long loops are in many cases susceptible to proteolytic degradation. One specific type of long loop, the omega loop, is compact with good internal packing interactions and is therefore quite stable. Other long loops, which by themselves would be attacked by proteolytic enzymes, are stabilized and protected by binding metal ions, especially calcium.

Schematic pictures of proteins highlight secondary structure

All pictorial representations of molecules are simplified versions of our current model of real molecules, which are quantum mechanical, probabilistic collections of atoms as both particles and waves. These are difficult to illustrate. Therefore we use different types of simplified representations, including space-filling models; ball-and-stick models, where atoms are spheres and bonds are sticks; and models that illustrate surface properties. The most detailed representation is the ball-and-stick model. However, a model of a protein structure where all atoms are displayed is confusing because of the sheer amount of information present (Figure 2.9a).

A two-dimensional picture of such a model is impossible to interpret. Even if the side-chain atoms are stripped off, it is still difficult to extract

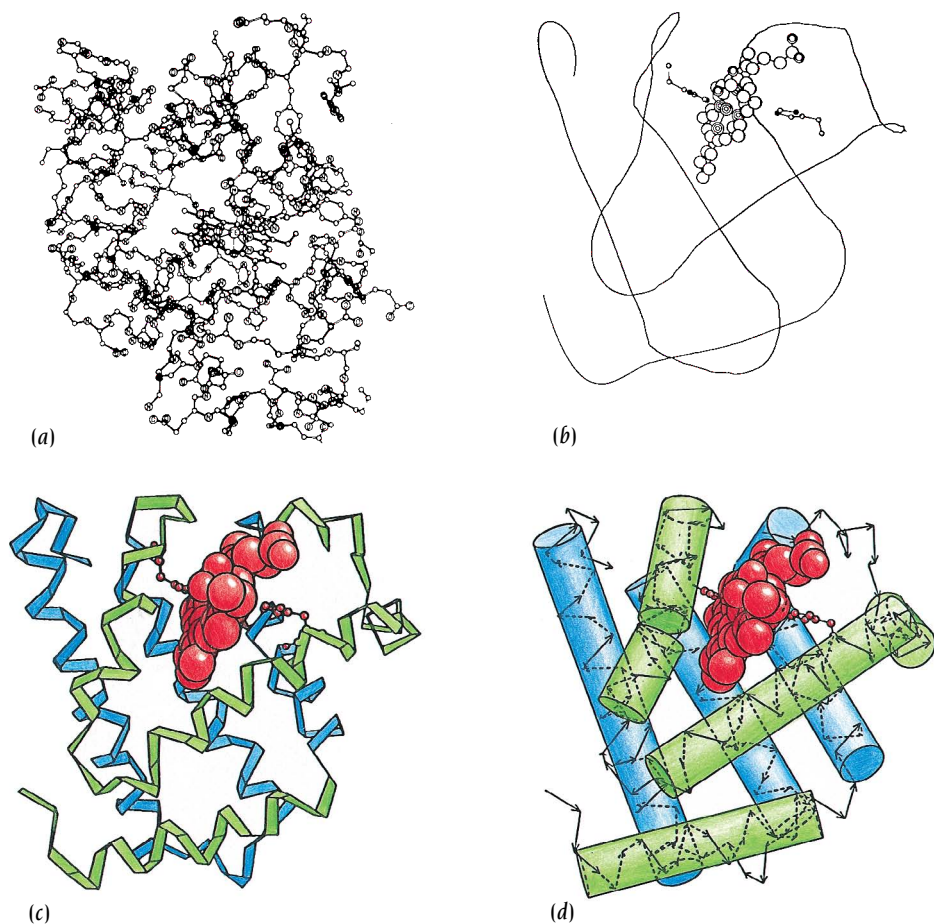
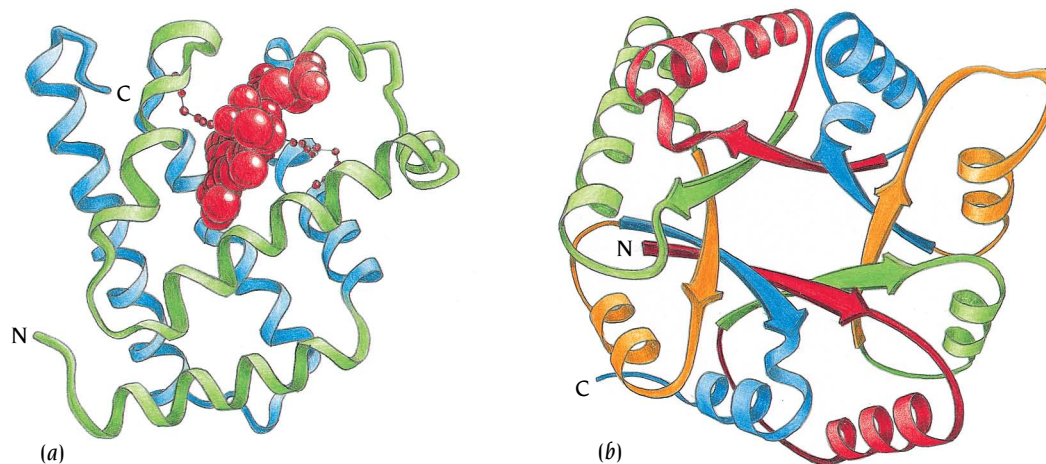


Figure 2.9 (a) The structure of myoglobin displaying all atoms as small circles connected by straight lines. Even though only side chains at the surface of the molecule are shown, the picture contains so many atoms that such a two-dimensional representation is very confusing and very little information can be gained from it. (b–d) Computer-generated schematic diagrams at different degrees of simplification of the structure of myoglobin. [(a) Half of a stereo diagram by H.C. Watson, *Prog. Stereochem.* 4: 299–333, 1969, by permission of Plenum Press. (b–d) From Arthur Lesk, *Protein Architecture: A Practical Approach*, 1991, Oxford University Press.]



meaningful information from a flat picture of such a model, mainly because it is difficult to see the relationships between the secondary structural elements. Since these elements dominate the structure, the picture becomes clearer if they are simplified and highlighted in some way. This is usually done by representing the path of the polypeptide chain by three different symbols: cylinders for α helices; arrows for β strands, which give the direction of the strands from amino to carboxy end; and ribbons for the remaining parts. Such schematic diagrams give good and very useful overall views of protein structures but, of course, give no detailed information. Details are best studied on graphic display systems where one can manipulate the computer-generated model on the screen; in this way the power of the graphics device makes it possible for the viewer to study much greater detail intelligibly.

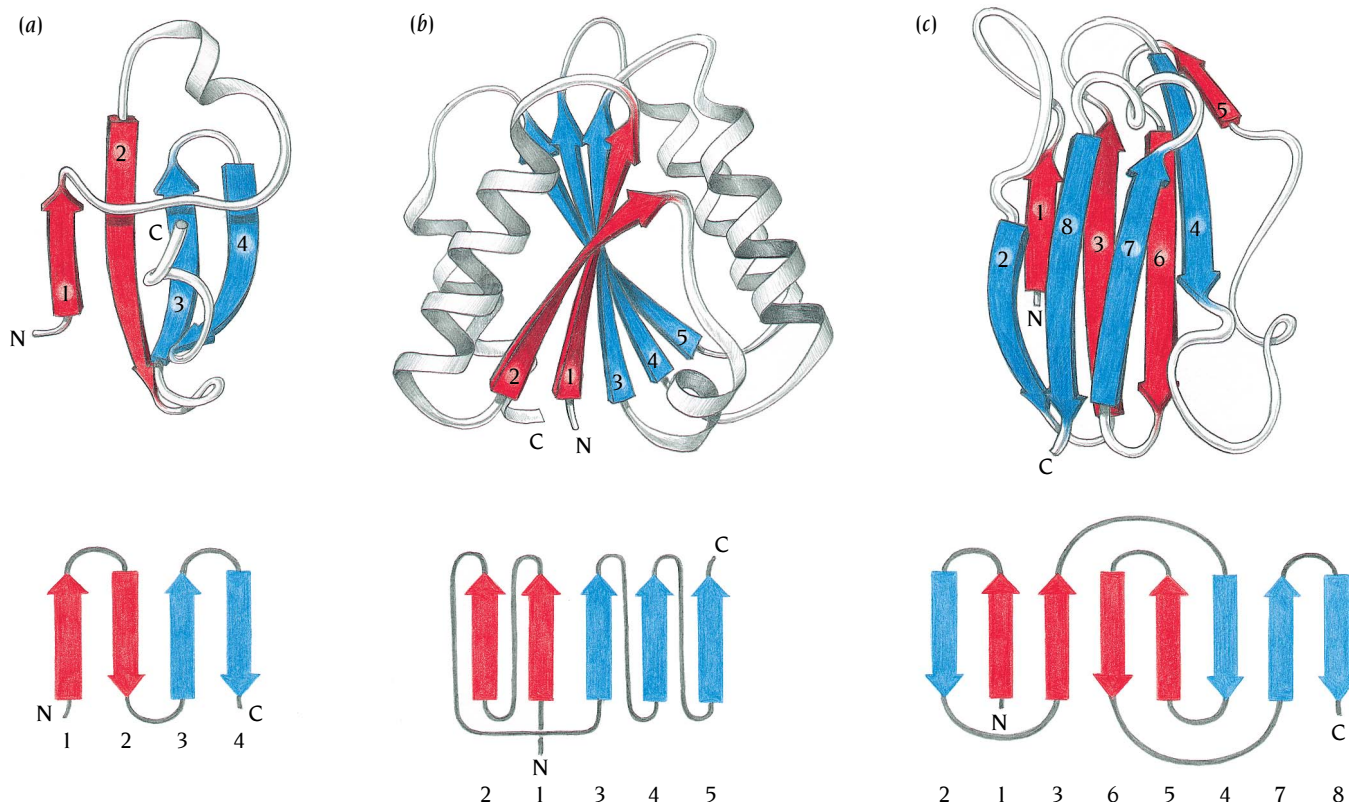
The Kinemage Supplement, produced by Jane Richardson at Duke University, is available from the publisher as a complement to this book for readers with access to a personal computer. The system is easy to use and provides interactive three-dimensional viewing of many of the structures discussed in various chapters.

Jane Richardson has also made a very popular collection of **schematic diagrams** of various protein molecules with an artistic touch that gives an aesthetic impression without losing too much accuracy. Arthur Lesk at the MRC Laboratory of Molecular Biology in Cambridge, UK, and Karl Hardman at IBM pioneered the use of computer programs to generate schematic diagrams on a computer display from a list of atomic coordinates of the main-chain atoms. Figures 2.9b–d and 2.10 show representative examples of both Lesk-type and Richardson-type diagrams.

Topology diagrams are useful for classification of protein structures

It is very convenient for some purposes to have an even more simplified schematic representation of the secondary structure elements, especially for β sheets. The most characteristic features of a β sheet are the number of strands, their relative directions (parallel or antiparallel), and how the strands are connected along the polypeptide chain (the strand order). This information can be easily conveyed through simple diagrams of connected arrows like those in Figure 2.11, where such simple **topology diagrams** are compared to the more elaborate Richardson diagrams. The twist of the β sheet is not represented in these topology diagrams. They are, nevertheless, very helpful when used to compare β structures and to analyze and present data in computerized database searches of similar structures. Such topology diagrams will be used frequently in this book.

Figure 2.10 Examples of schematic diagrams of the type pioneered by Jane Richardson. Diagram (a) illustrates the structure of myoglobin in the same orientation as the computer-drawn diagrams of Figures 2.9b–d. Diagram (b), which is adapted from J. Richardson, illustrates the structure of the enzyme triosephosphate isomerase, determined to 2.5 Å resolution in the laboratory of David Phillips, Oxford University. Such diagrams can easily be obtained from databases of protein structures, such as PDB, SCOP or CATH, available on the World Wide Web.



Secondary structure elements are connected to form simple motifs

Simple combinations of a few secondary structure elements with a specific geometric arrangement have been found to occur frequently in protein structures. These units have been called either supersecondary structures or **motifs**. We will use the term “motif” throughout this book. Some of these motifs can be associated with a particular function such as DNA binding; others have no specific biological function alone but are part of larger structural and functional assemblies.

The simplest motif with a specific function consists of two α helices joined by a loop region. Two such motifs, each with its own characteristic geometry and amino acid sequence requirements, have been observed as parts of many protein structures (Figure 2.12).

One of these motifs, called the helix-turn-helix motif, is specific for DNA binding and is described in detail in Chapters 8 and 9. The second motif is specific for calcium binding and is present in parvalbumin, calmodulin, troponin-C, and other proteins that bind calcium and thereby regulate cellular activities. This calcium-binding motif was first found in 1973 by Robert Kretsinger, University of Virginia, when he determined the structure of parvalbumin to 1.8 Å resolution.

Parvalbumin is a muscle protein with a single polypeptide chain of 109 amino acids. Its function is uncertain, but calcium binding to this protein probably plays a role in muscle relaxation. The helix-loop-helix motif appears three times in this structure, in two of the cases there is a calcium-binding site. Figure 2.13 shows this motif which is called an **EF hand** because the fifth and sixth helices from the amino terminus in the structure of parvalbumin, which were labeled E and F, are the parts of the structure that were originally used to illustrate calcium binding by this motif. Despite this trivial origin, the name has remained in the literature.

Figure 2.11 Beta sheets are usually represented simply by arrows in topology diagrams that show both the direction of each β strand and the way the strands are connected to each other along the polypeptide chain. Such topology diagrams are here compared with more elaborate schematic diagrams for different types of β sheets. (a) Four strands. Antiparallel β sheet in one domain of the enzyme aspartate transcarbamoylase. The structure of this enzyme has been determined to 2.8 Å resolution in the laboratory of William Lipscomb, Harvard University. (b) Five strands. Parallel β sheet in the redox protein flavodoxin, the structure of which has been determined to 1.8 Å resolution in the laboratory of Martha Ludwig, University of Michigan. (c) Eight strands. Antiparallel barrel in the electron carrier plastocyanin. This is a closed barrel where the sheet is folded such that β strands 2 and 8 are adjacent. The structure has been determined to 1.6 Å resolution in the laboratory of Hans Freeman in Sydney, Australia. (Adapted from J. Richardson.)

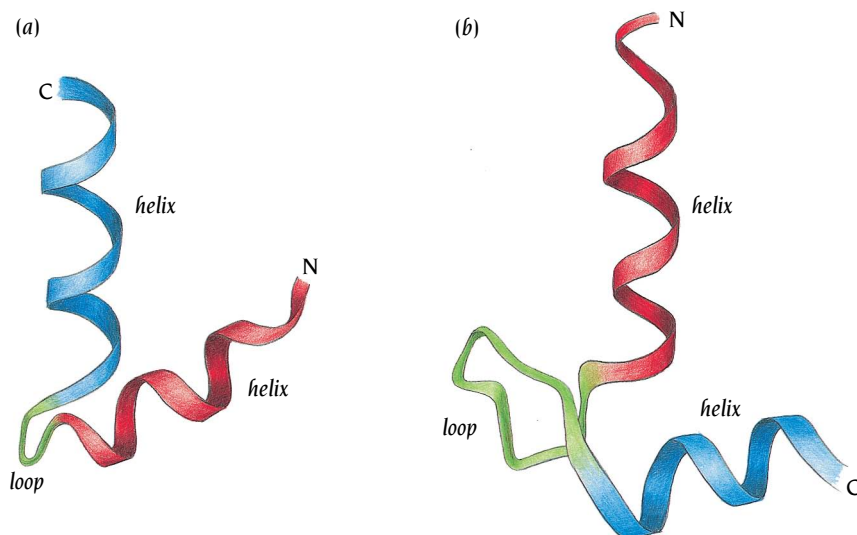


Figure 2.12 Two α helices that are connected by a short loop region in a specific geometric arrangement constitute a helix-turn-helix motif. Two such motifs are shown: the DNA-binding motif (a), which is further discussed in Chapter 8, and the calcium-binding motif (b), which is present in many proteins whose function is regulated by calcium.

The loop region between the two α helices binds the calcium atom. Carboxyl side chains from Asp and Glu, main-chain $C'=O$ and H_2O form the ligands to the metal atom (see Figure 2.13b). Thus both the specific main-chain conformation of the loop and specific side chains are required to provide the function of this motif. The helix-loop-helix motif provides a scaffold that holds the calcium ligands in the proper position to bind and release calcium.

When Kretsinger analyzed in detail the structure of the calcium-binding motif in parvalbumin in 1973, he deduced a set of constraints that an amino acid sequence must conform to in order to form such a motif, one of the first to be recognized in protein structures. The motif comprises two α helices, E and F, that flank a loop of 12 contiguous residues. Five of the loop residues are calcium ligands, and their side chains should contain an oxygen atom and preferably be Asp or Glu (Table 2.2). Residue 6 of the loop must be a glycine because the side chain of any other residue would disturb the structure of the motif. Finally, a number of side chains form a hydrophobic core between the α helices and thus must be hydrophobic. Applying these constraints, Kretsinger predicted that several different calcium-binding proteins could form this motif. Among these proteins were the important muscle

Figure 2.13 Schematic diagrams of the calcium-binding motif. (a) The calcium-binding motif is symbolized by a right hand. Helix E (red) runs from the tip to the base of the forefinger. The flexed middle finger corresponds to the green loop region of 12 residues that binds calcium (pink). Helix F (blue) runs to the end of the thumb. (b) The calcium atom is bound to one of the motifs in the muscle protein troponin-C through six oxygen atoms: one each from the side chains of Asp (D) 9, Asn (N) 11, and Asp (D) 13; one from the main chain of residue 15; and two from the side chain of Glu (E) 20. In addition, a water molecule (W) is bound to the calcium atom. (c) Schematic diagram illustrating that the structure of troponin-C is built up from four EF motifs—colored as in (a). Two of these bind Ca (pink balls) in the molecules that were used for the structure determination. (Adapted from a diagram by J. Richardson in O. Herzberg and M. James, *Nature* 313: 653–659, 1985.)

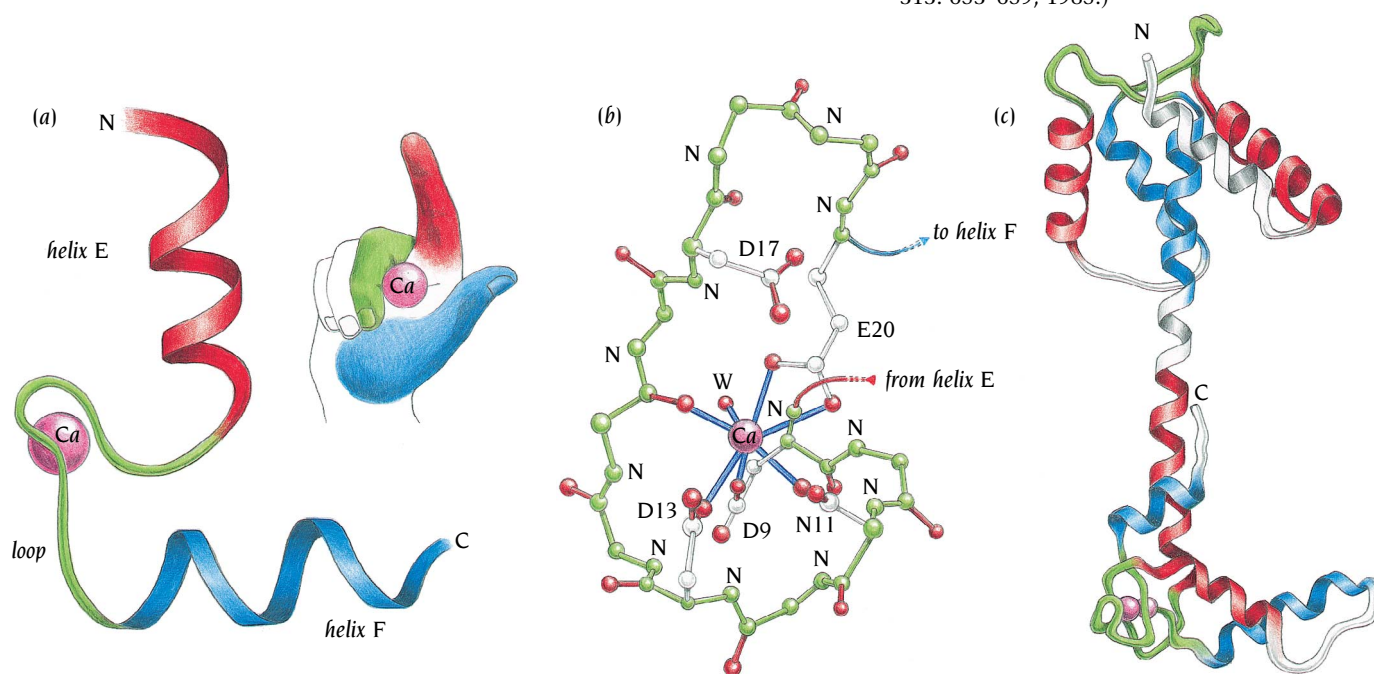


Table 2.2 Amino acid sequences of calcium-binding EF motifs in three different proteins

Parvalbumin	V	K	K	A	F	A	I	I	D	Q	D	K	S	G	F	I	E	E	D	E	L	K	L	F	L	Q	N	F
Calmodulin	F	K	E	A	F	S	L	F	D	K	D	G	D	G	T	I	T	T	K	E	L	G	T	V	M	R	S	L
Troponin-C	L	A	D	C	F	R	I	F	D	K	N	A	D	G	F	I	D	I	E	E	L	G	E	I	L	R	A	T



Calcium-binding residues are orange, and residues that form the hydrophobic core of the motif are light green. The helix-loop-helix region shown underneath is colored as in Figure 2.13.

protein troponin-C as well as calmodulin, which regulates a variety of cellular functions by calcium binding. Subsequent structure determinations of these two proteins have shown that Kretsinger's prediction was correct. The structure of one of these, troponin-C, which was determined by Osnat Herzberg in the laboratory of Michael James in Edmonton, Canada, to 2.0 Å resolution, is shown in Figure 2.13. Functional aspects of calcium binding to calmodulin and the dramatic structural change of this molecule when it binds to target peptides are discussed in Chapter 6.

The hairpin β motif occurs frequently in protein structures

The simplest motif involving β strands, simpler than the α -helical calcium-binding motif, is two adjacent antiparallel strands joined by a loop. This motif, which is called either a hairpin or a β - β unit, occurs quite frequently; it is present in most antiparallel β structures both as an isolated ribbon and as part of more complex β sheets. There is a strong preference for β strands to be adjacent in β sheets when they are adjacent in the amino acid sequence and thus to form a **hairpin β motif** (β hairpin for short). The lengths of the loop regions between the β strands vary but are generally from two to five residues long (see Figure 2.8). There is no specific function associated with this motif.

Figure 2.14 shows examples of both cases, an isolated ribbon and a β sheet. The isolated ribbon is illustrated by the structure of bovine trypsin inhibitor (Figure 2.14a), a small, very stable polypeptide of 58 amino acids that inhibits the activity of the digestive protease trypsin. The structure has been determined to 1.0 Å resolution in the laboratory of Robert Huber in Munich, Germany, and the folding pathway of this protein is discussed in Chapter 6. Hairpin motifs as parts of a β sheet are exemplified by the structure of a snake venom, erabutoxin (Figure 2.14b), which binds to and inhibits

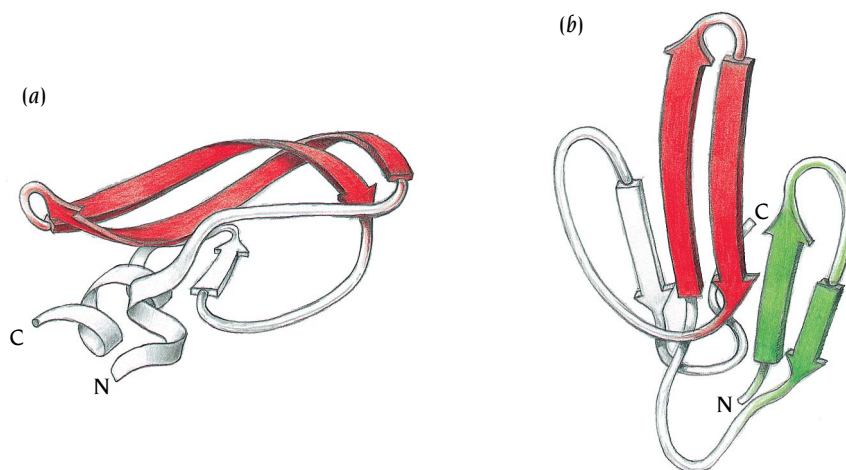


Figure 2.14 The hairpin motif is very frequent in β sheets and is built up from two adjacent β strands that are joined by a loop region. Two examples of such motifs are shown. (a) Schematic diagram of the structure of bovine trypsin inhibitor. The hairpin motif is colored red. (b) Schematic diagram of the structure of the snake venom erabutoxin. The two hairpin motifs within the β sheet are colored red and green. (Adapted from J. Richardson.)

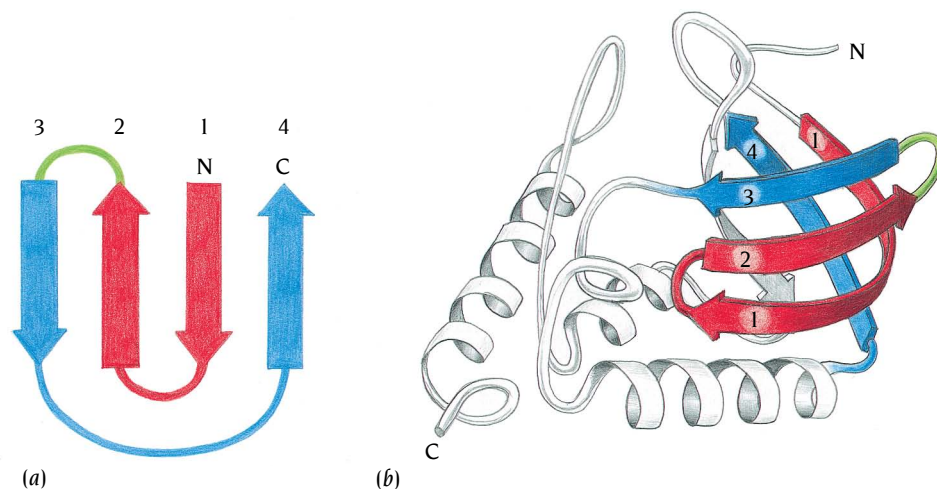


Figure 2.15 The Greek key motif is found in antiparallel β sheets when four adjacent β strands are arranged in the pattern shown as a topology diagram in (a). The motif occurs in many β sheets and is exemplified here by the enzyme *Staphylococcus* nuclease (b). The four β strands that form this motif are colored red and blue. The structure of this enzyme was determined to 1.5 Å resolution in the laboratory of Al Cotton at MIT. (Adapted from J. Richardson.)

the acetylcholine receptor in nerve cells. The structure has been determined to 1.4 Å resolution in the laboratory of Barbara Low at Columbia University. The core of this structure is a β sheet of five strands that contains two hairpin motifs and one additional β strand.

The Greek key motif is found in antiparallel β sheets

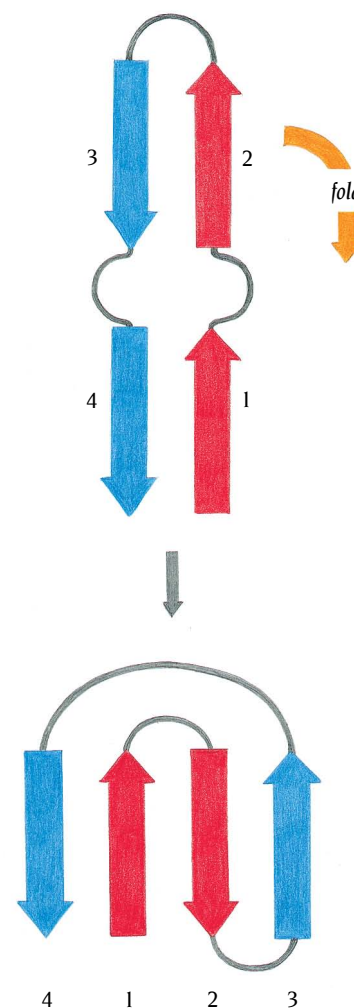
Four adjacent antiparallel β strands are frequently arranged in a pattern similar to the repeating unit of an ornamental pattern, or fret, used in ancient Greece, which is now called a Greek key. In proteins the motif is therefore called a **Greek key motif**, and Figure 2.15 shows an example of such a motif in the structure of *Staphylococcus* nuclease, an enzyme that degrades DNA. The Greek key motif is not associated with any specific function, but it occurs frequently in protein structures.

The Greek key motifs have been analyzed in detail, and it has been suggested that their frequent occurrence compared with other arrangements of four antiparallel β strands is based on an initial formation of one long antiparallel structure with loops in the middle of both β strands, as shown in Figure 2.16. By structural changes in the loop regions between β strands 1 and 2 and between β strands 3 and 4, the top part folds down so that β strand 2 associates with β strand 1. Beta strands 1 and 2 then form hydrogen bonds, and the Greek key motif is thus formed.

The β - α - β motif contains two parallel β strands

The hairpin motif is a simple and frequently used way to connect two antiparallel β strands, since the connected ends of the β strands are close together at the same edge of the β sheet. How are parallel β strands connected? If two adjacent strands are consecutive in the amino acid sequence, the two ends that must be joined are at opposite edges of the β sheet. The polypeptide chain must cross the β sheet from one edge to the other and connect the next β strand close to the point where the first β strand started. Such crossover connections are frequently made by α helices. The polypeptide chain must turn twice using loop regions, and the motif that is formed is thus a β strand followed by a loop, an α helix, another loop, and, finally, the second β strand.

Figure 2.16 Suggested folding pathway from a hairpinlike structure to the Greek key motif. Beta strands 2 and 3 fold over such that strand 2 is aligned adjacent and antiparallel to strand 1. The topology diagram of the Greek key shown here is the same as in Figure 2.15a but rotated 180° in the plane of the page.



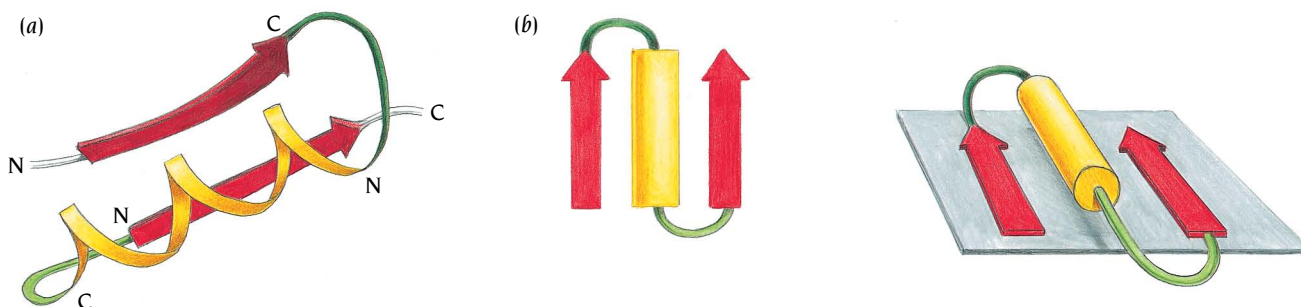


Figure 2.17 Two adjacent parallel β strands are usually connected by an α helix from the C-terminus of strand 1 to the N-terminus of strand 2. Most protein structures that contain parallel β sheets are built up from combinations of such β - α - β motifs. Beta strands are red, and α helices are yellow. Arrows represent β strands, and cylinders represent helices. (a) Schematic diagram of the path of the main chain. (b) Topological diagrams of the β - α - β motif.

This motif is called a **beta-alpha-beta motif** (Figure 2.17) and is found as part of almost every protein structure that has a parallel β sheet. For example, the molecule shown in Figure 2.10b, triosephosphate isomerase, is entirely built up by repeated combinations of this motif, where two successive motifs share one β strand. Alternatively, it can be regarded as being built up from four consecutive β - α - β - α motifs.

The α helix in the β - α - β motif connects the carboxy end of one β strand with the amino end of the next β strand (see Figure 2.17) and is usually oriented so that the helical axis is approximately parallel to the β strands. The α helix packs against the β strands and thus shields the hydrophobic residues of the β strands from the solvent. The β - α - β motif thus consists of two parallel β strands, an α helix, and two loop regions (except in a few cases where the connection between the two parallel β strands is not an α helix but a polypeptide chain of irregular structure). The loop regions can be of very different lengths, from one or two residues to over a hundred. The two loops have different functions. The loop (dark green in Figure 2.17) that connects the carboxy end of the β strand with the amino end of the α helix is often involved in forming the functional binding site, or active site, of these structures. These loop regions thus usually have conserved amino acid sequences in homologous proteins. In contrast, the other loop (light green in Figure 2.17) has not yet been found to contribute to an active site.

The β - α - β motif can be regarded as a loose helical turn from one β strand, around the connection, and into the next β strand. The motif can thus in principle have two different “hands” (Figure 2.18). Essentially every β - α - β motif in the known protein structures has been found to have the same hand as a right-handed α helix and therefore is called right-handed. No convincing explanation has been found for this regularity, even though it is the only general rule that describes how three secondary structure elements are arranged relative to each other. This handedness has important structural and functional consequences when several of these motifs are linked into a domain structure, as will be described in Chapter 4.

Protein molecules are organized in a structural hierarchy

The Danish biochemist Kai Linderstrøm-Lang coined the terms “primary,” “secondary,” and “tertiary” structure to emphasize the structural hierarchy in

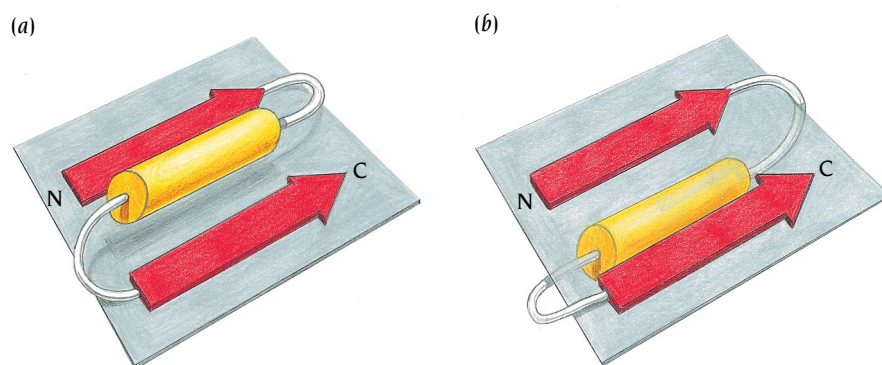


Figure 2.18 The β - α - β motif can in principle have two “hands.” (a) This connection with the helix above the sheet is found in almost all proteins and is called right-handed because it has the same hand as a right-handed α helix. (b) The left-handed connection with the helix below the sheet.

proteins (see Figure 1.1). **Primary structure** is the amino acid sequence, or, in other words, the arrangement of amino acids along a linear polypeptide chain. Two different proteins that have significant similarities in their primary structures are said to be homologous to each other, and since their corresponding DNA sequences also are significantly similar, it is generally assumed that the two proteins are evolutionarily related, that they have evolved from a common ancestral gene.

Secondary structure occurs mainly as α helices and β strands. The formation of secondary structure in a local region of the polypeptide chain is to some extent determined by the primary structure. Certain amino acid sequences favor either α helices or β strands; others favor formation of loop regions. Secondary structure elements usually arrange themselves in simple motifs, as described earlier. Motifs are formed by packing side chains from adjacent α helices or β strands close to each other.

Several motifs usually combine to form compact globular structures, which are called **domains**. In this book we will use the term **tertiary structure** as a common term both for the way motifs are arranged into domain structures and for the way a single polypeptide chain folds into one or several domains. In all cases examined so far it has been found that if there is significant amino acid sequence homology in two domains in different proteins, these domains have similar tertiary structures.

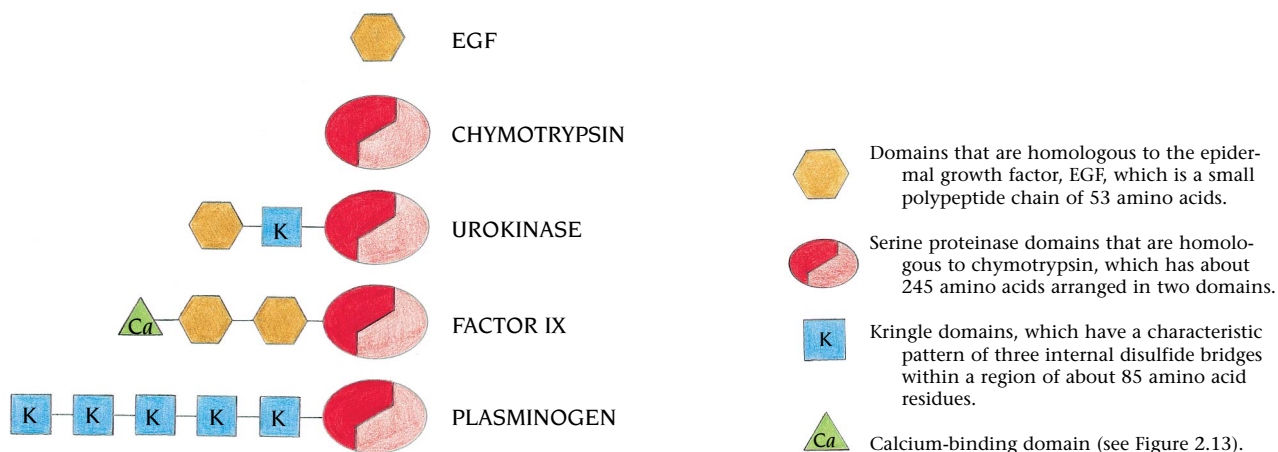
Protein molecules that have only one chain are called **monomeric** proteins. But a fairly large number of proteins have a **quaternary structure**, which consists of several identical polypeptide chains (subunits) that associate into a **multimeric** molecule in a specific way. These subunits can function either independently of each other or cooperatively so that the function of one subunit is dependent on the functional state of other subunits. Other protein molecules are assembled from several different subunits with different functions; for example, RNA polymerase from *E. coli* contains five different polypeptide chains.

Large polypeptide chains fold into several domains

The fundamental unit of tertiary structure is the domain. A domain is defined as a polypeptide chain or a part of a polypeptide chain that can fold independently into a stable tertiary structure. Domains are also units of function. Often, the different domains of a protein are associated with different functions. For example, in the lambda repressor protein, discussed in Chapter 8, one domain at the N-terminus of the polypeptide chain binds DNA, while a second domain at the C-terminus contains a site necessary for the dimerization of two polypeptide chains to form the dimeric repressor molecule.

Proteins may comprise a single domain or as many as several dozen domains (Figure 2.19). There is no fundamental structural distinction

Figure 2.19 Organization of polypeptide chains into domains. Small protein molecules like the epidermal growth factor, EGF, comprise only one domain. Others, like the serine proteinase chymotrypsin, are arranged in two domains that are required to form a functional unit (see Chapter 11). Many of the proteins that are involved in blood coagulation and fibrinolysis, such as urokinase, factor IX, and plasminogen, have long polypeptide chains that comprise different combinations of domains homologous to EGF and serine proteinases and, in addition, calcium-binding domains and Kringle domains.



between a domain and a subunit, there are many known examples where several biological functions that are carried out by separate polypeptide chains in one species are performed by domains of a single protein in another species. For example, synthesis of fatty acids requires catalysis of seven different chemical reactions. In plant chloroplasts these reactions are catalyzed by seven different proteins, whereas in mammals they are performed by one polypeptide chain arranged in seven domains with short linker regions between the domains. Such differences thus reflect the organization of the genome rather than the dictates of structure.

Domains are built from structural motifs

Domains are formed by different combinations of secondary structure elements and motifs. The α helices and β strands of the motifs are adjacent to each other in the three-dimensional structure and connected by loop regions. Sequentially adjacent motifs, or motifs that are formed from consecutive regions of the primary structure of a polypeptide chain, are usually close together in the three-dimensional structure (Figure 2.20). Thus to a first approximation a polypeptide chain can be considered as a sequential arrangement of these simple motifs. The number of such combinations found in proteins is limited, and some combinations seem to be structurally favored. Thus similar domain structures frequently occur in different proteins with different functions and with completely different amino acid sequences.

Simple motifs combine to form complex motifs

Figure 2.21 illustrates the 24 possible ways in which two adjacent β hairpin motifs, each consisting of two antiparallel β strands connected by a loop region, can be combined to make a more complex motif.

A survey of all known structures in 1991 showed that only those eight arrangements shown in Figure 2.21a occurred either as a complete β sheet or as a fragment of a β sheet with more than four strands. The number of times that these complex motifs occurred were 65, 29, 23, 11, 9, 3, 2, 1 for (i) to

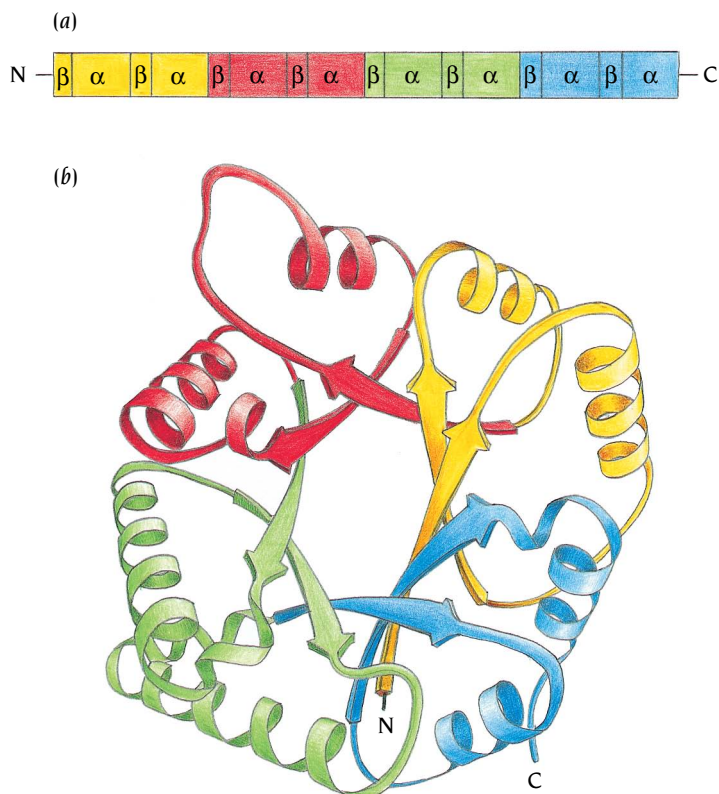


Figure 2.20 Motifs that are adjacent in the amino acid sequence are also usually adjacent in the three-dimensional structure. Triose-phosphate isomerase is built up from four β - α - β - α motifs that are consecutive both in the amino acid sequence (a) and in the three-dimensional structure (b).

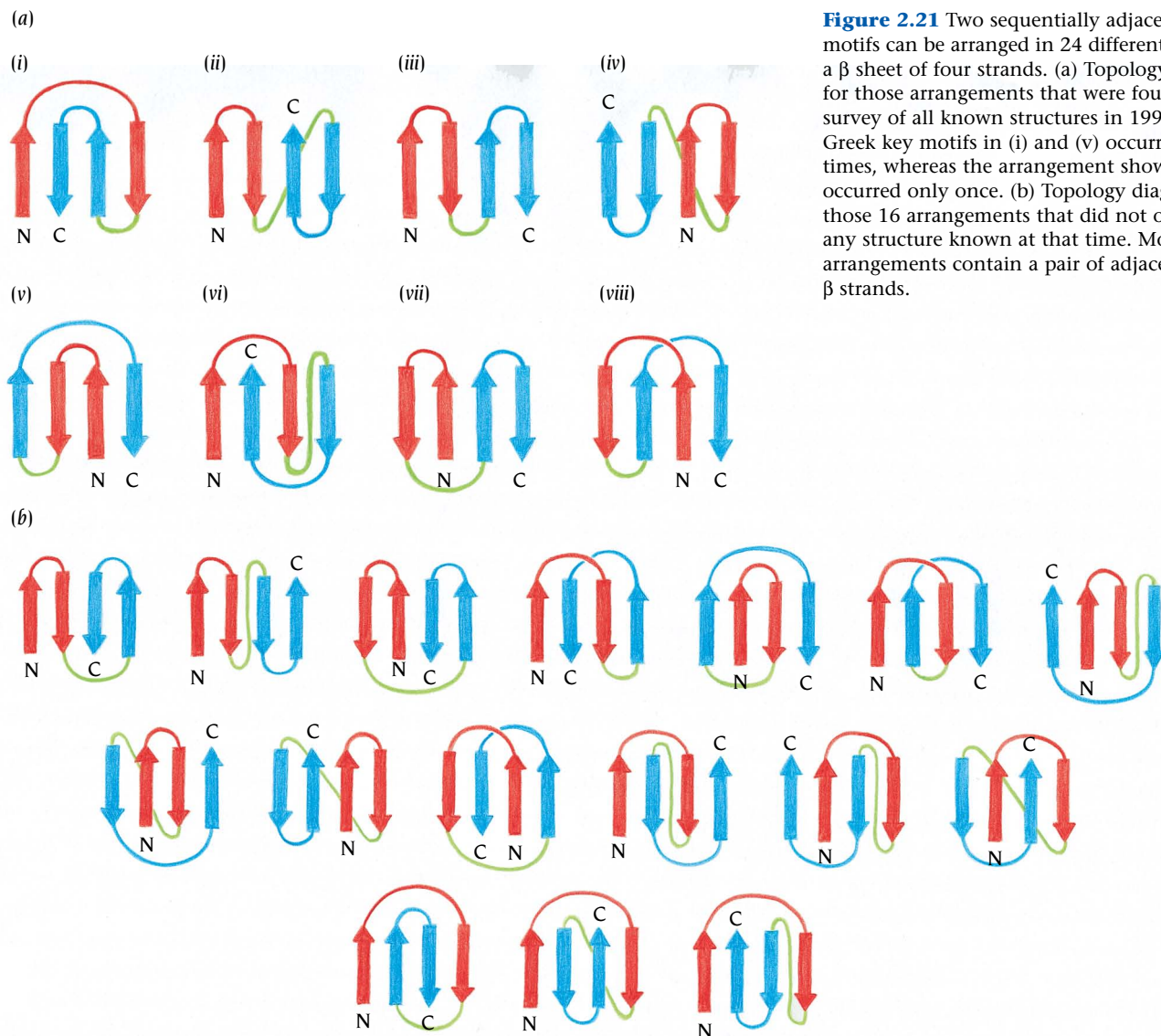


Figure 2.21 Two sequentially adjacent hairpin motifs can be arranged in 24 different ways into a β sheet of four strands. (a) Topology diagrams for those arrangements that were found in a survey of all known structures in 1991. The Greek key motifs in (i) and (v) occurred 74 times, whereas the arrangement shown in (viii) occurred only once. (b) Topology diagrams for those 16 arrangements that did not occur in any structure known at that time. Most of these arrangements contain a pair of adjacent parallel β strands.

(viii), respectively. The Greek key motifs shown in (i) and (v) thus occur more frequently than all the other possible motifs together. It is also apparent that two hairpin motifs strongly prefer to combine in such a way that all four strands become antiparallel, rather than being arranged with two adjacent parallel β strands. Of 143 β sheets with 4 strands, 138 were antiparallel and only 5 contained 2 adjacent parallel β strands.

The 16 possible motifs shown in Figure 2.21b never occur. Even if this database is limited compared with the universe of existing proteins, the survey clearly demonstrates that a few topological arrangements occur much more frequently than others, and that most possible complex motifs never occur or occur only in a few cases. In 1995 a preliminary survey of a much larger database of known structures yielded the same basic conclusions.

Protein structures can be divided into three main classes

On the basis of simple considerations of connected motifs, Michael Levitt and Cyrus Chothia of the MRC Laboratory of Molecular Biology derived a taxonomy of protein structures and have classified domain structures into three main groups: α domains, β domains, and α/β domains. In α structures the core is built up exclusively from α helices (see Figure 2.9); in β structures the core comprises antiparallel β sheets and are usually two β sheets packed

against each other (see Figure 2.11c). The α/β structures are made from combinations of β - α - β motifs that form a predominantly parallel β sheet surrounded by α helices (see Figures 2.10b and 2.11b).

Some proteins are built up from a combination of discrete α and β motifs and usually form one small antiparallel β sheet in one part of the domain packed against a number of α helices (see Figure 2.15). These structures can be considered to belong to a small fourth group called $\alpha + \beta$. In addition to these groups, there are a number of small proteins that are rich in disulfide bonds or metal atoms and form a special group. The structures of these proteins seem to be strongly influenced by the presence of these metals or disulfides and often look like distorted versions of more regular proteins.

In this book the domains of the known protein structures are classified according to Levitt and Chothia's scheme. The three main classes— α , β , and α/β —will be examined in more detail in Chapters 3, 4, and 5. The group of Cyrus Chothia has constructed a database in which all available protein structures are arranged according to this hierarchy. Within each class the structures are arranged in superfamilies according to their tertiary structure and, within the superfamilies, in families according to function and sequence homology. This database is freely available on the World Wide Web.

Conclusion

The interiors of protein molecules contain mainly hydrophobic side chains. The main chain in the interior is arranged in secondary structures to neutralize its polar atoms through hydrogen bonds. There are two main types of secondary structure, α helices and β sheets. Beta sheets can have their strands parallel, antiparallel, or mixed.

Protein structures are built up by combinations of secondary structural elements, α helices, and β strands. These form the core regions—the interior of the molecule—and they are connected by loop regions at the surface. Schematic and simple topological diagrams where these secondary structure elements are highlighted are very useful and are frequently used. Alpha helices or β strands that are adjacent in the amino acid sequence are also usually adjacent in the three-dimensional structure. Certain combinations, called motifs, occur very frequently, including the helix-loop-helix motif and the hairpin motif. A DNA-binding helix-loop-helix motif and a calcium-binding helix-loop-helix motif, each with its own specific geometry and amino acid sequence requirements, are used in many different proteins.

The β - α - β motif, which consists of two parallel β strands joined by an α helix, occurs in almost all structures that have a parallel β sheet. Four antiparallel β strands that are arranged in a specific way comprise the Greek key motif, which is frequently found in structures with antiparallel β sheets.

Polypeptide chains are folded into one or several discrete units, domains, which are the fundamental functional and three-dimensional structural units. The cores of domains are built up from combinations of small motifs of secondary structure, such as α -loop- α , β -loop- β , or β - α - β motifs. Domains are classified into three main structural groups: α structures, where the core is built up exclusively from α helices; β structures, which comprise antiparallel β sheets; and α/β structures, where combinations of β - α - β motifs form a predominantly parallel β sheet surrounded by α helices.

Selected readings

General

- Chothia, C. Principles that determine the structure of proteins. *Annu. Rev. Biochem.* 53: 537–572, 1984.
- Doolittle, R.F. Proteins. *Sci. Am.* 253: 88–99, 1985.
- Hardie, D.G., Coggins, J.R. *Multidomain Proteins: Structure and Evolution*. Amsterdam: Elsevier, 1986.
- Janin, J., Chothia, C. Domains in proteins: definitions, location and structural principles. *Methods Enzymol.* 115: 420–430, 1985.
- Klotz, I.M., et al. Quaternary structure of proteins. *Annu. Rev. Biochem.* 39: 25–62, 1970.
- Lesk, A.M. Themes and contrasts in protein structures. *Trends Biochem. Sci.* 9: June V, 1984.
- Levitt, M., Chothia, C. Structural patterns in globular proteins. *Nature* 261: 552–558, 1976.
- Matthews, B.W., Bernhard, S.A. Structure and symmetry of oligomeric enzymes. *Annu. Rev. Biophys. Bioeng.* 2: 257–317, 1973.
- Richardson, J.S. Describing patterns of protein tertiary structure. *Methods Enzymol.* 115: 349–358, 1985.
- Richardson, J.S. Schematic drawings of protein structures. *Methods Enzymol.* 115: 359–380, 1985.
- Richardson, J.S. The anatomy and taxonomy of protein structure. *Adv. Prot. Chem.* 34: 167–339, 1981.
- Rossmann, M.G., Argos, P. Protein folding. *Annu. Rev. Biochem.* 50: 497–532, 1981.
- Schulz, G.E. Protein differentiation: emergence of novel proteins during evolution. *Angew. Chem., int. ed.* 20: 143–151, 1981.
- Schulz, G.E. Structural rules for globular proteins. *Angew. Chem., int. ed.* 16: 23–33, 1977.
- Strynadka, N.C.J., James, M.N.G. Crystal structures of the helix-loop-helix calcium-binding proteins. *Annu. Rev. Biochem.* 58: 951–998, 1989.

Specific structures

- Adams, M.J., et al. Structure of lactate dehydrogenase at 2.8 Å resolution. *Nature* 227: 1098–1103, 1970.
- Baba, Y.S., et al. Three-dimensional structure of calmodulin. *Nature* 315: 37–40, 1985.
- Banner, B.W., et al. Structure of chicken muscle triose phosphate isomerase determined crystallographically at 2.5 Å resolution using amino acid sequence data. *Nature* 255: 609–614, 1975.
- Bourne, P.E., et al. Erabutoxin b. Initial protein refinement and sequence analysis at 0.140 nm resolution. *Eur. J. Biochem.* 153: 521–527, 1985.
- Burnett, R.M., et al. The structure of oxidized form of clostridial flavodoxin at 1.9 Å resolution. *J. Biol. Chem.* 249: 4383–4392, 1974.
- Chothia, C. Structural invariants in protein folding. *Nature* 254: 304–308, 1975.

- Chothia, C., Levitt, M., Richardson, D. Structure of proteins: packing of α -helices and pleated sheets. *Proc. Natl. Acad. Sci. USA* 74: 4130–4134, 1977.
- Chou, P.Y., Fasman, G.D. β -turns in proteins. *J. Mol. Biol.* 115: 135–175, 1977.
- Colman, P., et al. X-ray crystal structure analysis of plastocyanin at 2.7 Å resolution. *Nature* 272: 319–324, 1978.
- Crawford, J.L., Lipscomb, W.N., Schellmann, C.G. The reverse turn as a polypeptide conformation in globular proteins. *Proc. Natl. Acad. Sci. USA* 70: 538–542, 1973.
- Efimov, A.V. Stereochemistry of α -helices and β -sheet packing in compact globule. *J. Mol. Biol.* 134: 23–40, 1979.
- Eklund, H., et al. Three-dimensional structure of horse liver alcohol dehydrogenase at 2.4 Å resolution. *J. Mol. Biol.* 102: 27–59, 1976.
- Gouaux, J.E., Lipscomb, W.N. Crystal structures of phosphonoacetamide ligated T and phosphonoacetamide and malonate ligated R states of aspartate carbamoyltransferase at 2.8 Å resolution and neutral pH. *Biochemistry* 29: 389–402, 1990.
- Herzberg, O., James, M.N.G. Structure of the calcium regulatory muscle protein troponin-C at 2.8 Å resolution. *Nature* 313: 653–659, 1985.
- Hol, W.G.J., van Duijnen, P.T., Berendsen, H.J.C. The α -helix dipole and the properties of proteins. *Nature* 273: 443–446, 1978.
- Holmgren, A., et al. Three-dimensional structure of *E. coli* thioredoxin-S₂ to 2.8 Å resolution. *Proc. Natl. Acad. Sci. USA* 72: 2305–2309, 1975.
- Jones, A., Thirup, S. Using known substructures in protein model building and crystallography. *EMBO J.* 5: 819–822, 1986.
- Kendrew, J.C. The three-dimensional structure of a protein molecule. *Sci. Am.* 205: 96–110, 1961.
- Kendrew, J.C., et al. A three-dimensional model of the myoglobin molecule obtained by x-ray analysis. *Nature* 181: 662–666, 1958.
- Kendrew, J.C., et al. Structure of myoglobin. *Nature* 185: 422–427, 1960.
- Koch, I., Kaden, F., Selbig, J. Analysis of protein sheet topologies by graph theoretical methods. *Prot. Struct. Func. Genet.* 12: 314–323, 1992.
- Kretsinger, R.H. Structure and evolution of calcium-modulated proteins. *CRC Crit. Rev. Biochem.* 8: 119–174, 1980.
- Lesk, A.M., Hardman, K.D. Computer-generated pictures of proteins. *Methods Enzymol.* 115: 381–390, 1985.
- Levitt, M. Conformational preferences of amino acids in globular proteins. *Biochemistry* 17: 4277–4285, 1978.
- Matthews, B.W., Rossmann, M.G. Comparison of protein structures. *Methods Enzymol.* 115: 397–420, 1985.

- Milner-White, E.J., Poet, R. Loops, bulges, turns and hairpins in proteins. *Trends Biochem. Sci.* 12: 189–192, 1987.
- Moews, P.C., Kretsinger, R.H. Refinement of the structure of carp muscle calcium-binding parvalbumin by model building and difference Fourier analysis. *J. Mol. Biol.* 91: 201–228, 1975.
- Park, C.H., Tulinsky, A. Three-dimensional structure of the Kringle sequence: structure of prothrombin fragment 1. *Biochemistry* 25: 3977–3982, 1986.
- Pauling, L., Corey, R.B. Configurations of polypeptide chains with favored orientations around single bonds: two new pleated sheets. *Proc. Natl. Acad. Sci. USA* 37: 729–740, 1951.
- Pauling, L., Corey, R.B., Branson, H.R. The structure of proteins: two hydrogen-bonded helical configurations of the polypeptide chain. *Proc. Natl. Acad. Sci. USA* 37: 205–211, 1951.
- Perutz, M.F. Electrostatic effects in proteins. *Science* 201: 1187–1191, 1978.
- Perutz, M.F. New x-ray evidence on the configuration of polypeptide chains. Polypeptide chains in poly-g-benzyl-t-glutamate, keratin and haemoglobin. *Nature* 167: 1053–1054, 1951.
- Perutz, M.F., et al. Structure of haemoglobin. A three-dimensional Fourier synthesis at 5.5 Å resolution, obtained by x-ray analysis. *Nature* 185: 416–422, 1960.
- Rao, S.T., Rossmann, M.G. Comparison of super-secondary structures in proteins. *J. Mol. Biol.* 76: 241–256, 1973.
- Remington, S.J., Matthews, B.W. A systematic approach to the comparison of protein structures. *J. Mol. Biol.* 140: 77–99, 1980.
- Richards, F.M. Calculation of molecular volumes and areas for structures of known geometry. *Methods Enzymol.* 115: 440–464, 1985.
- Rose, G.D. Automatic recognition of domains in globular proteins. *Methods Enzymol.* 115: 430–440, 1985.
- Rose, G.D. Prediction of chain turns in globular proteins on a hydrophobic basis. *Nature* 272: 586–590, 1978.
- Rose, G.D., Roy, S. Hydrophobic basis of packing in globular proteins. *Proc. Natl. Acad. Sci. USA* 77: 4643–4647, 1980.
- Rose, G.D., Young, W.B., Gierasch, L.M. Interior turns in globular proteins. *Nature* 304: 654–657, 1983.
- Sibanda, B.L., Thornton, J.M. β -hairpin families in globular proteins. *Nature* 316: 170–174, 1985.
- Tucker, P.W., Hazen, E.E., Cotton, F.A. Staphylococcal nuclease reviewed: a prototypic study in contemporary enzymology. III. Correlation of the three-dimensional structure with the mechanisms of enzymatic action. *Mol. Cell. Biochem.* 23: 67–86, 1979.
- Venkatachalam, C.M. Stereochemical criteria for polypeptides and proteins. V. Conformation of a system of three linked peptide units. *Biopolymers* 6: 1425–1436, 1968.
- Wiegand, G., et al. Crystal structure analysis and molecular model of a complex of citrate synthase with oxaloacetate and S-acetyl-coenzyme A. *J. Mol. Biol.* 174: 205–219, 1984.
- Wlodawer, A., Deisenhofer, J., Huber, R. Comparison of two highly refined structures of bovine pancreatic trypsin inhibitor. *J. Mol. Biol.* 193: 145–156, 1987.
- Wright, C.S., Alden, R., Kraut, J. Structure of subtilisin BPN' at 2.5 Å resolution. *Nature* 221: 235–242, 1969.

The first globular protein structure that was determined, myoglobin, belongs to the class of alpha- (α -) domain structures. The structure illustrated in Figure 2.9 is called the globin fold and is a representative example of one class of α domains in proteins; short α helices, the building blocks, are connected by loop regions and packed together to produce a hydrophobic core. Packing interactions within the core hold the helices together in a stable globular structure, while the hydrophilic residues on the surface make the protein soluble in water. In this chapter we will describe some of the different α -domain structures in soluble proteins.

Alpha helices are sufficiently versatile to produce many very different classes of structures. In membrane-bound proteins, the regions inside the membranes are frequently α helices whose surfaces are covered by hydrophobic side chains suitable for the hydrophobic environment inside the membranes. Membrane-bound proteins are described in Chapter 12. Alpha helices are also frequently used to produce structural and motile proteins with various different properties and functions. These can be typical fibrous proteins such as keratin, which is present in skin, hair, and feathers, or parts of the cellular machinery such as fibrinogen or the muscle proteins myosin and dystrophin. These α -helical proteins will be discussed in Chapter 14.

Coiled-coil α helices contain a repetitive heptad amino acid sequence pattern

Despite its frequent occurrence in proteins an isolated α helix is only marginally stable in solution. Alpha helices are stabilized in proteins by being packed together through hydrophobic side chains. The simplest way to achieve such stabilization is to pack two α helices together. As early as 1953 Francis Crick showed that the side-chain interactions are maximized if the two α helices are not straight rods but are wound around each other in a supercoil, a so-called **coiled-coil** arrangement (Figure 3.1). Coiled-coils are the basis for some of the fibrous proteins we shall discuss in Chapter 14. Coiled-coils in fibers can extend over many hundreds of amino acid residues

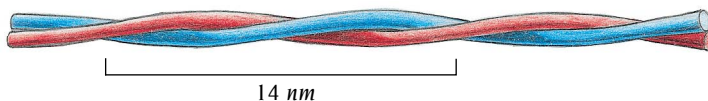


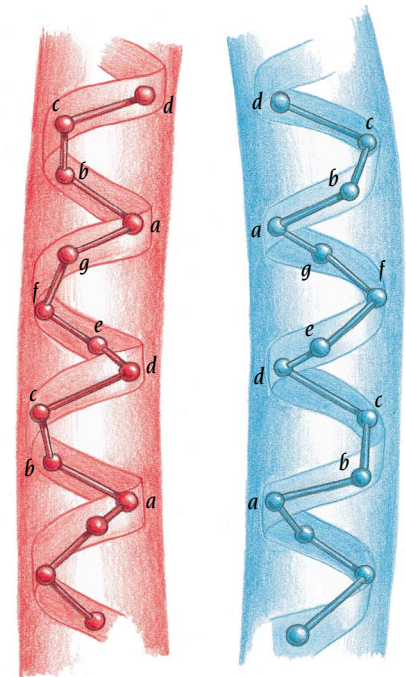
Figure 3.1 Schematic diagram of the coiled-coil structure. Two α helices are intertwined and gradually coil around each other.

NH_2 - *a* - *b* - *c* - *d* - *e* - *f* - *g* -
 Met - Lys - Gln - **Leu** - Glu - Asp - Lys -
 Val - Glu - Glu - **Leu** - Leu - Ser - Lys -
 Asn - Tyr - His - **Leu** - Glu - Asn - Glu -
 Val - Ala - Arg - **Leu** - Lys - Lys - Leu - COOH

(a)

Figure 3.2 Repetitive pattern of amino acids in a coiled-coil α helix. (a) The amino acid sequence of the transcription factor GCN4 showing a heptad repeat of leucine residues. Within each heptad the amino acids are labeled a–g. (b) Schematic diagram of one heptad repeat in a coiled-coil structure showing the backbone of the polypeptide chain. The α helices in the coiled-coil are slightly distorted so that the helical repeat is 3.5 residues rather than 3.6, as in a regular helix. There is therefore an integral repeat of seven residues along the helix.

(b)



to produce long, flexible dimers that contribute to the strength and flexibility of the fibers. Much shorter coiled-coils are used in some transcription factors to promote or prevent formation of homo- and heterodimers, as we shall discuss in Chapter 10.

Crick showed that a left-handed supercoil of two right-handed α helices reduces the number of residues per turn in each helix from 3.6 to 3.5 so that the pattern of side-chain interactions between the helices repeats every seven residues, that is, after two turns. This is reflected in the amino acid sequences of polypeptide chains that form α -helical coiled-coils. Such sequences are repetitive with a period of seven residues, the **heptad repeat**. The amino acid residues within one such heptad repeat are usually labeled a–g (Figure 3.2a), and one of these, the d-residue, is hydrophobic, usually a leucine or an isoleucine. When two α helices form a coiled-coil structure the side chains of these d-residues pack against each other every second turn of the α helices (Figure 3.2b). The hydrophobic region between the α helices is completed by the a-residues, which are frequently hydrophobic and also pack against each other (Figure 3.3). Residues “e” and “g,” which border the hydrophobic core (see Figure 3.2b), frequently are charged residues. The side chains of these residues provide ionic interactions (salt bridges) between the α helices that define the relative chain alignment and orientation (Figure 3.4).

The repetitive heptad amino acid sequence pattern required for a coiled-coil structure can be identified in computer searches of amino acid sequence databases. Heptad repeats provide strong indications of α -helical coiled-coil structures, and they have been found in a number of different proteins with very diverse functions. Fibrinogen, which plays an essential role in blood coagulation; some RNA- and DNA-binding proteins; the class of cell-surface recognition proteins called collectins; both spectrin and dystrophin, which link actin molecules; and the muscle protein myosin all contain heptad repeats and therefore coiled-coil α helices. An illustrative example is provided by GCN4, a DNA-binding protein. GCN4 contains one region of α helix, the leucine zipper region, and its dimerization is accomplished by the formation of an α -helical coiled-coil with the leucine zipper regions of two subunits. The structure and DNA-binding function of this protein are described in Chapter 10.

Detailed structure determinations of GCN4 and other coiled-coil proteins have shown that the α helices pack against each other according to the “knobs in holes” model first suggested by Francis Crick (Figure 3.5). Each side chain in the hydrophobic region of one of the α helices can contact four side chains from the second α helix. The side chain of a residue in position “d”

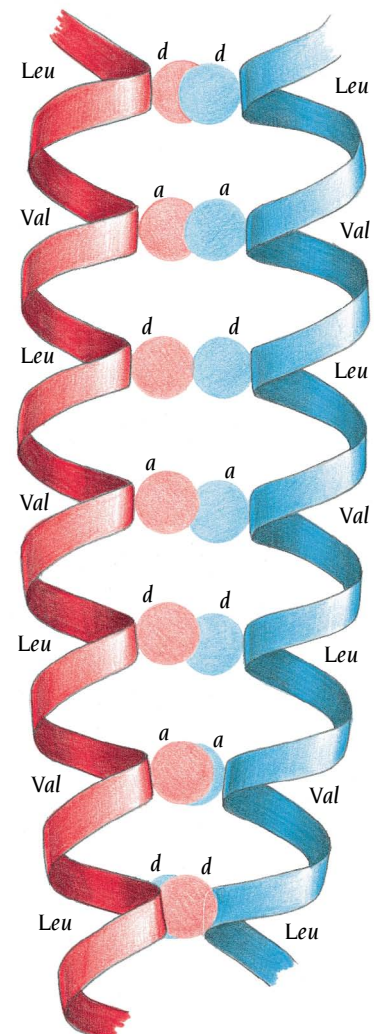


Figure 3.3 Schematic diagram showing the packing of hydrophobic side chains between the two α helices in a coiled-coil structure. Every seventh residue in both α helices is a leucine, labeled “d.” Due to the heptad repeat, the d-residues pack against each other along the coiled-coil. Residues labeled “a” are also usually hydrophobic and participate in forming the hydrophobic core along the coiled-coil.

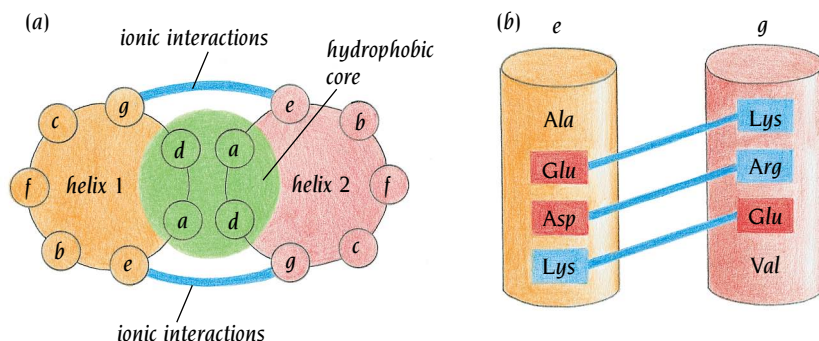


Figure 3.4 Salt bridges can stabilize coiled-coil structures and are sometimes important for the formation of heterodimeric coiled-coil structures. The residues labeled “e” and “g” in the heptad sequence are close to the hydrophobic core and can form salt bridges between the two α helices of a coiled-coil structure, the e-residue in one helix with the g-residue in the second and vice versa. (a) Schematic view from the top of a heptad repeat. (b) Schematic view from the side of a coiled-coil structure.

in one helix is directed into a hole at the surface of the second helix surrounded by one d-residue, two a-residues, and one e-residue, with numbers n , $n - 3$, $n + 4$, and $n + 1$, respectively. The two helices are aligned in such a way that the two d-residues, frequently leucines or isoleucines, face each other (see Figure 3.3).

The four-helix bundle is a common domain structure in α proteins

Two α helices packed together into a coiled-coil are building blocks within a domain or a fiber but are not sufficient to form a complete domain. The simplest and most frequent α -helical domain consists of four α helices arranged in a bundle with the helical axes almost parallel to each other. A schematic representation of the structure of the **four-helix bundle** is shown in Figure 3.6a. The side chains of each helix in the four-helix bundle are arranged so that hydrophobic side chains are buried between the helices and hydrophilic side chains are on the outer surface of the bundle (Figure 3.6b). This arrangement creates a hydrophobic core in the middle of the bundle along its length, where the side chains are so closely packed that water is excluded.

The four-helix bundle occurs in several widely different proteins, such as myohemerythrin (an oxygen-transport protein in marine worms that does not contain heme iron), cytochrome c' and cytochrome b₅₆₂ (heme-containing electron carriers) (Figure 3.7a), ferritin (a storage molecule for iron atoms in eucaryotic cells), and the coat protein of tobacco mosaic virus. In these examples, sequentially adjacent α helices are always antiparallel. However, four-helix bundles can also be formed with different topological arrangements of the α helices. In human growth hormone (Figure 3.7b), a four-helix bundle is formed from two pairs of parallel α helices that are joined in an

Figure 3.5 Schematic diagram of packing side chains in the hydrophobic core of coiled-coil structures according to the “knobs in holes” model. The positions of the side chains along the surface of the cylindrical α helix is projected onto a plane parallel with the helical axis for both α helices of the coiled-coil. (a) Projected positions of side chains in helix 1. (b) Projected positions of side chains in helix 2. (c) Superposition of (a) and (b) using the relative orientation of the helices in the coiled-coil structure. The side-chain positions of the first helix, the “knobs,” superimpose between the side-chain positions in the second helix, the “holes.” The green shading outlines a d-residue (leucine) from helix 1 surrounded by four side chains from helix 2, and the brown shading outlines an a-residue (usually hydrophobic) from helix 1 surrounded by four side chains from helix 2.

